

PROOF OF THE TROPICAL HODGE CONJECTURE FOR TROPICAL ABELIAN VARIETIES

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ABSTRACT. In every ambient dimension, for every full-rank real period lattice, and in every cycle dimension, we prove that the rational classes in the kernel of the eigenwave on a flat integral affine translation torus are exactly the classes of finite signed balanced straight tropical cycles with rational coefficients. Representatives have the exact prescribed marked homology class on the original torus, with bounded rational-faced cells and finitely many cell orbits. In particular, every rational tropical Hodge class on every tropical abelian variety has such a representative. The proof uses complete endpoint moments, finite signed specialization and filling, periodic polyhedral realization, and a direct Stokes calculation of the full class. The main result of this paper was obtained using the Danus system.

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1. INTRODUCTION

Tropical homology assigns a second degree to the homology of a tropical space by retaining exterior tangent coefficients. This construction makes it possible to compare balanced polyhedral cycles with Hodge-theoretic conditions. We study this comparison on flat integral affine translation tori. The question is whether the linear eigenwave condition on a rational diagonal class is sufficient for an actual finite balanced cycle with that class.

Let N be a free abelian group of rank $d \geq 1$, and put $V = N \otimes_{\mathbb{Z}} \mathbb{Q}$ and $V_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}$. Let $\Gamma \subset V_{\mathbb{R}}$ be any full-rank discrete lattice, set $G = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, and write $X = V_{\mathbb{R}}/\Gamma$. The tangent lattice and the period lattice have different roles. The former determines rational directions and primitive tangent vectors; the latter determines the torus and its homology marking. The period vectors may have arbitrary real coordinates in a basis of N . In this setting diagonal tropical homology is

$$H_{r,r}(X; \mathbb{Q}) = H_r(X; \bigwedge^r V) = \bigwedge^r G \otimes_{\mathbb{Q}} \bigwedge^r V.$$

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For $1 \leq r \leq d$, the real-target eigenwave on this rational group is the map

$$\phi_r : \bigwedge^r G \otimes_{\mathbb{Q}} \bigwedge^r V \longrightarrow \bigwedge_{\mathbb{R}}^{r-1} (G \otimes_{\mathbb{Q}} \mathbb{R}) \otimes_{\mathbb{R}} \bigwedge_{\mathbb{R}}^{r+1} V_{\mathbb{R}}$$

given by

$$\phi_r((g_1 \wedge \cdots \wedge g_r) \otimes \beta) = \sum_{a=1}^r (-1)^{a-1} (g_1 \wedge \cdots \widehat{g_a} \cdots \wedge g_r) \otimes (g_a \wedge \beta).$$

In the last factor g_a is its actual real period vector, and the tangent coefficient is extended to the real numbers. We set $\phi_0 = 0$. A rational tropical Hodge class is an element of $\ker \phi_r$. The rational domain is part of this definition, even when the target requires real coefficients.

A straight tropical cycle is locally a weighted balanced polyhedral complex with rational face directions. Here weights are rational and may have either sign. More precisely, upstairs on $V_{\mathbb{R}}$ we use a locally finite Γ -periodic face-to-face complex of bounded convex polytopes with finitely many cell orbits. At each codimension-one face, the sum of the incident weights times their inward primitive integral quotient normals is zero. Affine offsets may be real. The class map cl attaches the oriented primitive tangent exterior generator to each weighted cell and retains the full tensor displayed above. These are the rational straight cycles of the Mikhalkin–Zharkov framework [MZ14, Section 4.1, pp. 327–329].

1.1. Previous work and the scope of the question. The tropical homology theory of Itenberg, Katzarkov, Mikhalkin, and Zharkov [IKMZ19] supplies the bigraded framework. Mikhalkin and Zharkov construct the eigenwave and the canonical class of a straight cycle, and prove that such classes are killed by the eigenwave [MZ14, Section 5, Theorem 1, p. 334]. Their straight-cycle construction is the external class formalism used here; its precise version and locators are given in Section 6. Zharkov records Kontsevich’s proposal to test the classical Hodge conjecture by finding a tropical Hodge class on a degenerating abelian variety which has no balanced tropical representative [Zha20]. The necessary specialization comparison in that proposal is distinct from an assertion that every tropical cycle lifts to an algebraic cycle.

The converse to the eigenwave condition was already known in codimension one. Jell, Rau, and Shaw prove an integral tropical Lefschetz (1,1) theorem: on a tropical manifold, the kernel of the wave map on integral codimension-one Borel–Moore homology consists of tropical cycle classes [JRS18, Theorem 1.1]. Their result does not impose an additional rational-triangulability hypothesis. On a compact torus Borel–Moore and ordinary homology agree. Thus the codimension-one range of the present rational theorem is already covered by this stronger integral result.

Amini and Piquerez prove the all-codimension theorem for smooth projective tropical varieties admitting a rational triangulation [AP20a, Theorem 1.1]. Precisely, under those hypotheses, the locus of classes of rational codimension- p tropical cycles is the kernel of

$$N : H_{\text{trop}}^{p,p}(Y; \mathbb{Q}) \longrightarrow H_{\text{trop}}^{p-1,p+1}(Y; \mathbb{Q}).$$

Their argument uses a tropical Clemens–Schmid sequence and the Hodge-theoretic structures developed in [AP20b]. The paragraphs following their theorem expressly distinguish general tropical abelian varieties, which need not be rationally triangulable. Their general formulation keeps a rational source and a real monodromy target [AP20a, Conjecture 1.2]. Piquerez’s thesis records the theorem and that formulation in Chapter 1, Theorem 8.1 and Conjecture 8.2, and discusses the polarized flat-torus question in Section 8.4, Question 8.7 [Piq21, pp. 106, 109–110]. Irrational entries in a chosen period matrix alone are not a test for failure of rational triangulability.

The historical record also includes a later announcement. The official announcement of Amini’s talk at the IMJ-PRG algebraic geometry seminar on 20 June 2024 describes an outline of a proof of the tropical Hodge conjecture in joint work with Piquerez [Ami24]. Its abstract does not specify the complete hypotheses or coefficient conventions. The precise scope of any stronger

announced result has not been determined from a written theorem and proof available to the author. We therefore distinguish this announcement from the inspected written theorem just described, without inferring from the announcement either the removal or the retention of its rational-triangulability hypothesis. Qizheng Yin informed the author that the result had been announced at this seminar.

Several related developments supply further context. Jell, Shaw, and Smacka develop the superform description of tropical cohomology and its Poincaré duality [JSS19]. Amini and Piquerez’s work on fan homology and fan Hodge theory develops the local homological and Hodge-theoretic structures [AP21, AP23]; their subsequent work treats the tropical Feichtner–Yuzvinsky comparison, positivity criteria, homological smoothness, and Deligne resolutions [AP24a, AP24b]. Aksnes, Amini, Piquerez, and Shaw compare tropical and algebraic cohomology for cohomologically tropical varieties and in degenerations satisfying their stated hypotheses [AAPS25]. These comparisons address information beyond the ordinary cycle-class image of an abstract translation torus.

Amini, Corey, and Monin develop tropical Abel–Jacobi theory and the associated cycle and monodromy relations [ACM25]. Ghosh and Shokrieh construct the tropical Poincaré bundle and Fourier–Mukai transform and prove a generalized Poincaré formula [GS25]. Mikami constructs tropical cohomology by reductions; the hypothesis in his main comparison includes a unimodular $\mathbb{Q}$ -rational polyhedral complex structure [Mik26, Theorem 1.1]. These results concern related cohomological structures and comparisons, with the hypotheses stated in those works. Our proof addresses the cycle-realization problem directly for arbitrary real period lattices and rational cycle coefficients.

The counterexample strategy also has a classical geometric history. Brosnan’s maximal-degeneration restriction and Markman’s algebraicity results already exclude the indicated fourfold and sixfold Weil-class tests [Bro26, Mar23, Mar25]. We give their precise discriminant conventions, dimensional scopes, and relationship to the present theorem in Section 8. The question addressed by the proof below is the unrestricted ordinary tropical class image on a flat torus. A rational triangulation, a polarization, and a geometric origin in a specified degeneration are not its hypotheses.

1.2. The realization theorem. We show that the eigenwave condition suffices for a finite rational signed straight cycle on every flat translation torus. To state also the control of its vertices, fix a basis of N and let $S \subset \mathbb{R}$ be any finite-dimensional rational vector space containing the coordinates of a basis of Γ . Put $A = S \otimes_{\mathbb{Q}} V \subset V_{\mathbb{R}}$. Keeping the period coordinates in S gives the parameter-valued eigenwave

$$W_r : \bigwedge^r G \otimes \bigwedge^r V \longrightarrow \bigwedge^{r-1} G \otimes S \otimes \bigwedge^{r+1} V$$

with the same alternating formula. All tensors here are over $\mathbb{Q}$. Its rational-domain kernel equals $\ker \phi_r$, as the proof in Section 7 establishes.

Theorem 1.1. *For every rank $d \geq 1$ integral tangent lattice N , every full-rank real period lattice $\Gamma \subset V_{\mathbb{R}}$, and every $0 \leq r \leq d$, the complete marked homology classes of finite rational signed straight tropical r -cycles on $X = V_{\mathbb{R}}/\Gamma$ are exactly $\ker \phi_r$. For $1 \leq r \leq d$ this is precisely $\ker W_r$. Every class in this kernel has a representative with lifted vertices in $A = S \otimes V$ for every parameter space S specified above. It has finitely many cell orbits, bounded convex rational-faced cells, rational signed weights, primitive tangent framing, and ordinary balancing, on the original torus and in the exact given rational class. In degree zero the classes are represented by rational weighted points.*

This is Theorem 7.2. The period lattice is unrestricted apart from discreteness and full rank. The equality concerns the complete marked tensor, before any symmetrization, and the representative uses the original lattice. Rationality is required of the weights and face directions, while affine offsets may be real. The statement does not impose nonnegative weights.

A polarization is a homomorphism $\lambda : \Gamma \rightarrow N^* = \text{Hom}(N, \mathbb{Z})$ whose real extension gives a symmetric positive definite form $(\gamma, \delta) \mapsto \lambda(\gamma)(\delta)$ on $V_{\mathbb{R}}$. A translation torus with such a polarization is a tropical abelian variety. The realization theorem gives the following consequence.

Corollary 1.2. *For every tropical abelian variety of every dimension and every cycle dimension, each rational tropical Hodge class is the class of a finite rational signed straight tropical cycle.*

This is Corollary 7.3. The conclusion is a rational tropical Hodge theorem. It does not assert algebraic liftability of the representatives or the classical Hodge conjecture. For a concrete elementary instance, take $N = \mathbb{Z}$, $\Gamma = \sqrt{2}\mathbb{Z}$, and $X = \mathbb{R}/(\sqrt{2}\mathbb{Z})$. If $\gamma = \sqrt{2}$ denotes the positive period generator, the circle with rational weight c has marked class $c\gamma \otimes 1$. An interval from 0 to $\sqrt{2}$ gives one lifted cell, and its endpoints are identified by the original period. This realizes every rational degree-one class on this torus; degree-zero classes are rational weighted points.

1.3. The proof and its consequences. The proof combines finite incidence algebra with a geometric realization argument. Its first input is the complete endpoint moment module. Quotienting exterior-coefficient point chains by boundaries of incident segments retains a finite direct sum of exterior parameter moments, rather than only the coefficient sum. Translations act by finite nilpotent exponentials on this module. The resulting comparison of the Laurent Koszul differential with its linear differential lifts every eigenwave-kernel tensor over the original integral period group.

The second input is exactness of the incidence complex below its top degree. Parameter specialization is constructed as a finite signed chain operation, including folded and collapsed cells. Sweeping retains the entire exterior coefficient. At a junction, equal cap contractions can leave a nonzero horizontal cap residual; the filling argument retains that residual. Slicing and two well-founded inductions then give finite fillings in the original parameter space. Thus the method uses rational-faced polytopes with real offsets without requiring a rational triangulation of the torus.

The geometric step refines finitely many cells and their period translates into a genuine periodic polyhedral complex. The ordinary primitive framing identifies its incidence boundary with tropical balancing, in the straight-cycle class formalism of Mikhalkin–Zharkov [MZ14, Proposition 11, pp. 328–329]. Finally, a direct Euler–Stokes calculation identifies the full marked class. The iterated factors $1/q$ cancel the factor $r!$ from the alternating period expansion, leaving the explicit sign $(-1)^{r(r-1)/2}$, which is incorporated into the cycle. This calculation preserves the given class itself and the original deck lattice.

The all-dimensional, all-period conclusion also determines the ordinary tropical nonrepresentability test in Kontsevich’s proposal. Every candidate in the rational eigenwave kernel has a signed balanced representative. In particular, a linear class functional vanishing on all such cycles vanishes on every candidate Hodge class. Section 8 explains this consequence in detail, including Zharkov’s flag construction, the earlier Markman–Brosnan exclusions, and the distinction between ordinary representability and the finer specialization or liftability data of a chosen degeneration.

Section 2 fixes the incidence and class conventions. Section 3 proves the complete endpoint-moment and period-lifting results. Section 4 constructs specialization, caps, and sweeps, and Section 5 proves the finite filling theorems. Section 6 constructs periodic refinements and identifies the resulting geometric cycle classes. Section 7 completes the Stokes calculation and proves the realization theorem and its abelian corollary. Section 8 analyzes the counterexample strategy and the exact boundary of its exclusion.

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details of the use of generative AI are documented in Appendix A for transparency. The author is fully responsible for the correctness of the paper. The author would like to thank Qizheng Yin for useful discussions. The author would like to thank Ruochuan Liu and Gang Tian for constant support and encouragement.

2. NOTATION AND TERMINOLOGIES

Set-up 2.1. Let $d \geq 1$, let N be a free abelian group of rank d , and set $V = N \otimes_{\mathbb{Z}} \mathbb{Q}$ and $V_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}$. Let $\Gamma \subset V_{\mathbb{R}}$ be a full-rank discrete lattice, put $G = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, and put $X = V_{\mathbb{R}}/\Gamma$. Fix a basis of N and a finite-dimensional rational subspace $S \subset \mathbb{R}$ containing the coordinates of a basis of Γ . Set $A = S \otimes_{\mathbb{Q}} V \subset V_{\mathbb{R}}$ and denote the actual period inclusion by $j : G \rightarrow A$. All unspecified tensor products and exterior powers are over $\mathbb{Q}$; real exterior powers are indicated when necessary. The real extension $j_{\mathbb{R}}$ identifies $G \otimes_{\mathbb{Q}} \mathbb{R}$ with $V_{\mathbb{R}}$.

Definition 2.2. A *rational-faced polytope* in the real extension of a rational vector space U is a bounded convex polytope whose every face has tangent space $L \otimes_{\mathbb{Q}} \mathbb{R}$ for a rational subspace $L \subset U$. Its affine constants may be arbitrary real numbers. A *finite admissible subdivision* is an actual finite subdivision into such polytopes, with vertices in the specified parameter-valued position space and with induced orientations. Auxiliary ordinary triangulations used for integration need not be admissible subdivisions.

Definition 2.3. Let U be a finite-dimensional rational vector space, let $R \subset \mathbb{R}$ be a finite-dimensional rational subspace, and let $p \geq 0$. For $0 \leq q \leq p$, the space $C_{q,p}(R; U)$ is generated over $\mathbb{Q}$ by oriented rational-faced q -polytopes with vertices in $R \otimes U$. On a polytope of rational tangent space L , its coefficient belongs to

$$\bigwedge^q L \wedge \bigwedge^{p-q} U \subset \bigwedge^p U.$$

In degree zero this coefficient space is $\bigwedge^p U$. The relations are orientation reversal, coefficient linearity, and all finite admissible subdivisions. Boundary is the oriented facet boundary with the entire exterior coefficient retained. The groups outside $0 \leq q \leq p$ are zero. We write T_x for translation by $x \in R \otimes U$. For the standing torus, $C_{\bullet,r} = C_{\bullet,r}(S; V)$, and $(C_{\bullet,r})_{\Gamma}$ denotes its ordinary translation coinvariants, with relations $T_{\gamma}c - c$ for $\gamma \in \Gamma$.

Definition 2.4. A *finite rational signed straight tropical r -cycle* on X is a Γ -periodic locally finite face-to-face complex of bounded convex rational-faced polytopes, with finitely many cell orbits and rational weights on its r -cells. At every $(r-1)$ -face F it satisfies

$$\sum_{P \supset F} w_P u_{P/F} = 0 \quad \text{in } V/L_F,$$

where $u_{P/F}$ is a lift of the primitive integral quotient normal pointing from F into P , and L_F is the rational tangent space of F . Negative weights and arbitrary real affine offsets are allowed. Presentations are identified under finite compatible refinements and addition of coincident weights. If P has tangent L_P and orientation o , its tautological coefficient is $w_P \xi_{P,o}$, where $\xi_{P,o}$ is the oriented primitive generator of $\bigwedge^r(N \cap (L_P \otimes \mathbb{R}))$. The cycle class cl is the ordinary constant-exterior-coefficient class of this tautological chain, with the positive period marking

$$H_r(X; \bigwedge^r V) = \bigwedge^r G \otimes \bigwedge^r V.$$

The standard tropical homology notation for this group is $H_{r,r}(X; \mathbb{Q})$. Its rational *Hodge classes* are the kernel of the real eigenwave restricted to this rational group.

Definition 2.5. For $1 \leq r \leq d$, define the *parameter-valued eigenwave*

$$W_r : \bigwedge^r G \otimes \bigwedge^r V \rightarrow \bigwedge^{r-1} G \otimes S \otimes \bigwedge^{r+1} V$$

by

$$W_r((g_1 \wedge \cdots \wedge g_r) \otimes \beta) = \sum_{a=1}^r (-1)^{a-1} (g_1 \wedge \cdots \widehat{g_a} \cdots \wedge g_r) \otimes (j(g_a) \wedge \beta),$$

where $(s \otimes v) \wedge \beta = s \otimes (v \wedge \beta)$. Its real-target version uses the real period vectors and the corresponding real exterior spaces; its domain here remains the rational marked group. For $r = 0$ the eigenwave is zero. For $r = d$ its displayed target is zero.

Definition 2.6. A *polarization* on X is a homomorphism $\lambda : \Gamma \rightarrow N^* = \text{Hom}(N, \mathbb{Z})$ whose real extension defines a symmetric positive definite form $(\gamma, \delta) \mapsto \lambda(\gamma)(\delta)$ on $V_{\mathbb{R}}$. A translation torus equipped with such a polarization is called a *tropical abelian variety*.

3. ENDPOINT MOMENTS AND PERIOD LIFTS

We use the finite incidence chains of Definition 2.3. We first identify the quotient of endpoint chains by incident segment boundaries. Its translation action then gives an explicit group-homology lift for every class annihilated by the eigenwave.

Definition 3.1. Let U have rational dimension m , let $R \subset \mathbb{R}$ be a nonzero finite-dimensional rational subspace, and let $1 \leq p \leq m$. Set

$$M_p(R; U) = C_{0,p}(R; U) / \partial C_{1,p}(R; U), \quad E_p(R; U) = \bigoplus_{k=0}^{m-p} \bigwedge^k R \otimes \bigwedge^{p+k} U.$$

For $x = s \otimes u$ define $D_x(\alpha \otimes \beta) = (s \wedge \alpha) \otimes (u \wedge \beta)$, and extend linearly in $x \in R \otimes U$. Let $\epsilon : E_p(R; U) \rightarrow \bigwedge^p U$ be projection to degree zero.

We call k the *moment degree* of the summand $\bigwedge^k R \otimes \bigwedge^{p+k} U$. The operators D_x raise this degree by one. For pure tensors, exchanging two operators changes the sign in each exterior factor, so the operators commute. Every product of $m - p + 1$ such operators is zero. Thus

$$\exp(D_x) = \sum_{k=0}^{m-p} \frac{D_x^k}{k!}$$

is a finite polynomial. We identify $s \otimes u$ with the vector su in $U \otimes \mathbb{R}$.

Theorem 3.2. Let U, R, p be as in Definition 3.1. Then

$$[x] \otimes \beta \mapsto \exp(D_x)\beta$$

induces an isomorphism $M_p(R; U) \cong E_p(R; U)$, equivariant for translation by x acting as $\exp(D_x)$. An endpoint chain has zero image in this module if and only if it is the boundary of a finite incident segment chain with rational coefficients, rational directions, and vertices in the original space $R \otimes U$.

In particular, a finite chain $\sum_{\ell} [x_{\ell}] \otimes \beta_{\ell}$ bounds exactly when all its exterior moments vanish:

$$\sum_{\ell} \frac{D_{x_{\ell}}^k}{k!} \beta_{\ell} = 0 \quad \text{in} \quad \bigwedge^k R \otimes \bigwedge^{p+k} U, \quad 0 \leq k \leq m - p.$$

The degree-zero condition is the vanishing of the coefficient sum.

Proof. We present the segment relations, prove a finite nilpotence bound for translations, and use their logarithms to construct the inverse moment map. The presentation also gives the required finite segment filling.

Step 1. We present the endpoint module by its finite segment relations. Let $\mathcal{R} = \mathbb{Q}[R \otimes U]$ be the group algebra of the additive group $R \otimes U$. Its basis elements T_x act by translation, and we

put $\nabla_x = T_x - 1$. Identifying $[x] \otimes \beta$ with $T_x \otimes \beta$ gives

$$(3.1) \quad M_p(R; U) = \frac{\mathcal{R} \otimes \wedge^p U}{\left\langle \nabla_{su} \otimes (u \wedge \eta) : s \in R, u \in U, \eta \in \wedge^{p-1} U \right\rangle_{\mathcal{R}}}.$$

To verify this presentation, let a nonconstant admissible segment have endpoints x, y and rational direction $\mathbb{Q}u$. Choose a rational linear functional $\theta \in U^*$ with $\theta(u) = 1$. Since $y - x \in R \otimes U$, its scalar displacement $s = \theta(y - x)$ belongs to R , and $y - x = su$. Its coefficient is $u \wedge \eta$ for some $\eta \in \wedge^{p-1} U$, and its boundary is

$$\partial([x, x + su] \otimes (u \wedge \eta)) = T_x \nabla_{su} \otimes (u \wedge \eta).$$

Conversely, every translated generator in (3.1) is this boundary. The case $su = 0$ contributes zero, and finite segment subdivision telescopes to the same boundary. We write $\bar{\beta}$ for the class of $[0] \otimes \beta$ in $M_p(R; U)$.

Step 2. We prove a finite nilpotence bound for translation differences on the endpoint module. Choose a rational basis $e_1, \dots, e_m$ of U . Let $\mathfrak{a}_i$ be the ideal of $\mathcal{R}$ generated by ∇_{se_i} for $s \in R$. For $J = \{b_1 < \dots < b_p\}$, write $e_J = e_{b_1} \wedge \dots \wedge e_{b_p}$. If $i \in J$, then $\mathfrak{a}_i \bar{e}_J = 0$ by the segment relation.

We claim that $\mathfrak{a}_i^2 M_p(R; U) = 0$ for every i . It suffices to check the product $\nabla_{se_i} \nabla_{te_i} \bar{e}_J$. The case $i \in J$ has just been proved. Otherwise choose $b \in J$, which is possible because $p \geq 1$, and write $e_J = \pm e_b \wedge \eta$. The relation in the direction $e_i + e_b$ gives

$$\nabla_{t(e_i+e_b)} \overline{(e_i + e_b) \wedge \eta} = 0.$$

Multiply by ∇_{se_i} . Translation operators commute, and ∇_{se_i} kills $\overline{e_i \wedge \eta}$, so

$$\nabla_{se_i} \nabla_{t(e_i+e_b)} \overline{e_b \wedge \eta} = 0.$$

Now use

$$\nabla_{t(e_i+e_b)} = \nabla_{te_i} + \nabla_{te_b} + \nabla_{te_i} \nabla_{te_b}.$$

The terms containing ∇_{te_b} kill $\overline{e_b \wedge \eta}$. The remaining term is the claimed square-zero product. Since the $\bar{e}_J$ generate the module over $\mathcal{R}$, the claim follows.

Let $\mathfrak{a}$ be the augmentation ideal of $\mathcal{R}$ and put $\nu = m - p + 1$. Every $x \in R \otimes U$ has a unique expression $x = \sum_i s_i e_i$ with $s_i \in R$. The identity $T_x = \prod_i T_{s_i e_i}$ shows that $\mathfrak{a} = \sum_i \mathfrak{a}_i$. A product of ν coordinate ideals kills every $\bar{e}_J$: a repeated index gives a square-zero factor, while ν distinct indices must meet J , whose complement has size $m - p$. Hence

$$(3.2) \quad \mathfrak{a}^\nu M_p(R; U) = 0.$$

The same proof applies after every rational change of basis. Taking a nonzero $u \in U$ as a basis vector therefore also gives

$$\nabla_{su} \nabla_{tu} M_p(R; U) = 0 \quad (s, t \in R, u \in U).$$

For $u = 0$ this identity holds because $\nabla_0 = 0$.

Step 3. We construct translation logarithms and verify the two exterior alternations needed for the inverse moment map. On $M_p(R; U)$ define

$$L_x = \sum_{b=1}^{\nu-1} \frac{(-1)^{b+1}}{b} \nabla_x^b.$$

The sum is zero if $\nu = 1$. In view of (3.2), the polynomial logarithm and exponential identities modulo terms of total degree ν apply to these commuting operators. They give

$$L_{x+y} = L_x + L_y, \quad T_x = \exp(L_x).$$

The additive group $R \otimes U$ is a rational vector space, so $L_{qx} = qL_x$ for $q \in \mathbb{Q}$. Thus L_x is rational-linear in x . The square-zero identity on a rational line gives $L_{su} = \nabla_{su}$. The segment relation and its polarization in u now imply

$$(3.3) \quad \begin{aligned} L_{su} \overline{u \wedge \eta} &= 0, \\ L_{su} \overline{v \wedge \eta} &= -L_{sv} \overline{u \wedge \eta} \quad (u, v \in U, s \in R). \end{aligned}$$

For the second identity, expand the first one with $u + v$ in place of u ; its two diagonal terms vanish.

Consider the multilinear expression

$$L_{s_1 u_1} \cdots L_{s_k u_k} \overline{v_1 \wedge \cdots \wedge v_p}.$$

It is alternating in the coefficient slots $v_1, \dots, v_p$. To exchange an operator slot u_i with a coefficient slot v_b , commute the other logarithms aside, move v_b to the first position in the coefficient wedge, and apply (3.3). Restoring the coefficient order leaves exactly one minus sign. Since there is at least one coefficient slot, exchanges between operator slots are also products of three such exchanges through a coefficient slot. The expression is therefore alternating in all its $p + k$ slots from U .

Commuting two logarithms simultaneously exchanges their R -slots and their U -slots without changing the expression. The alternation just proved in the U -slots then forces alternation in the k slots from R . We consequently obtain well-defined maps

$$\Psi_k : \bigwedge^k R \otimes \bigwedge^{p+k} U \longrightarrow M_p(R; U)$$

given by

$$(3.4) \quad \begin{aligned} \Psi_k((s_1 \wedge \cdots \wedge s_k) \otimes (u_1 \wedge \cdots \wedge u_k \wedge v_1 \wedge \cdots \wedge v_p)) \\ = L_{s_1 u_1} \cdots L_{s_k u_k} \overline{v_1 \wedge \cdots \wedge v_p}. \end{aligned}$$

Their direct sum $\Psi : E_p(R; U) \rightarrow M_p(R; U)$ is surjective. Indeed, the quotient is spanned over $\mathbb{Q}$ by $T_x \bar{\beta} = \exp(L_x) \bar{\beta}$. Expanding x in the basis of U and β in exterior basis wedges expresses every term of this finite exponential as a value of one of the maps Ψ_k .

Step 4. We show that the moment map and the logarithmic construction are inverse equivariant maps. Define $\mu : C_{0,p}(R; U) \rightarrow E_p(R; U)$ by $\mu([x] \otimes \beta) = \exp(D_x) \beta$. Linearity of D_x and commutativity give $\mu T_y = \exp(D_y) \mu$. Moreover, $D_{su}^2 = 0$ and $D_{su}(u \wedge \eta) = 0$, whence

$$\mu(T_x \nabla_{su} \otimes (u \wedge \eta)) = \exp(D_x) (\exp(D_{su}) - \text{id})(u \wedge \eta) = 0.$$

Thus μ induces an equivariant map $\bar{\mu} : M_p(R; U) \rightarrow E_p(R; U)$. Applying the same finite logarithm to its translation-equivariance identity gives

$$\bar{\mu} L_x = D_x \bar{\mu}.$$

For a decomposable tensor in (3.4), the composition $\bar{\mu} \Psi_k$ is therefore

$$\begin{aligned} D_{s_1 u_1} \cdots D_{s_k u_k} (v_1 \wedge \cdots \wedge v_p) \\ = (s_1 \wedge \cdots \wedge s_k) \otimes (u_1 \wedge \cdots \wedge u_k \wedge v_1 \wedge \cdots \wedge v_p). \end{aligned}$$

These tensors span each summand, so $\bar{\mu} \Psi = \text{id}$. Together with the surjectivity of Ψ , this proves that $\bar{\mu}$ and Ψ are inverse isomorphisms.

Step 5. We realize each zero-moment endpoint chain as the boundary of finitely many admissible segments. If an endpoint chain has zero moment vector, the isomorphism and (3.1) express it as a finite sum

$$\sum_{\ell=1}^n c_\ell T_{x_\ell} \nabla_{s_\ell u_\ell} \otimes (u_\ell \wedge \eta_\ell), \quad c_\ell \in \mathbb{Q}, \quad x_\ell \in R \otimes U, \quad s_\ell \in R.$$

Here $u_\ell \in U$ and $\eta_\ell \in \bigwedge^{p-1} U$; finiteness follows because the generating submodule and the group algebra use finite sums. This is the boundary of the actual finite chain

$$\sum_{\ell=1}^n c_\ell [x_\ell, x_\ell + s_\ell u_\ell] \otimes (u_\ell \wedge \eta_\ell).$$

Every endpoint remains in $R \otimes U$. If a common segment complex is desired, split at the finitely many intersections and at endpoints of collinear overlaps. An isolated intersection of rational-direction lines is obtained by solving rational linear equations with constants in R , so its coordinates remain in R . These are finite admissible subdivisions. Conversely, the moment map vanishes on every segment boundary, as proved above. This gives both directions of the finite boundary criterion. $\square$

For an ordered integral basis $\delta_1, \dots, \delta_g$ of a free abelian subgroup of $R \otimes U$, we write $\delta_I = \delta_{i_1} \wedge \dots \wedge \delta_{i_a}$ when $I = \{i_1 < \dots < i_a\}$, and put $\delta_\emptyset = 1$. An empty alternating sum below denotes the zero map.

Theorem 3.3. *Let U, R, p be as in Definition 3.1, and let $\Delta \subset R \otimes U$ be a free abelian subgroup with a finite integral basis $\delta_1, \dots, \delta_g$. Put $H = \Delta \otimes \mathbb{Q}$ and $B = \bigwedge^p U$. For $0 \leq a \leq g$ let*

$$W_{a,p}((h_1 \wedge \dots \wedge h_a) \otimes \beta) = \sum_{i=1}^a (-1)^{i-1} (h_1 \wedge \dots \wedge \widehat{h_i} \wedge \dots \wedge h_a) \otimes D_{h_i} \beta.$$

Then the image of $H_a(\Delta; M_p(R; U)) \rightarrow H_a(\Delta; B) = \bigwedge^a H \otimes B$ induced by coefficient sum is exactly $\ker W_{a,p}$. More precisely, put

$$F_i = \sum_{b=0}^{m-p} \frac{D_{\delta_i}^b}{(b+1)!}.$$

For $y = \sum_{|I|=a} \delta_I \otimes \beta_I \in \ker W_{a,p}$, the expression

$$L_a(y) = \sum_{|I|=a} \delta_I \otimes \left(\prod_{i \in I} F_i^{-1} \right) \beta_I$$

is a cycle of the Koszul complex whose differential uses $\exp(D_{\delta_i}) - 1$, and its degree-zero coefficient sum is y . All inverses and sums are finite. The group acting throughout is Δ .

Under Theorem 3.2, coefficient sum is exactly ϵ : the degree-zero term of $\exp(D_x)\beta$ is β . The formula for L_a supplies the higher-moment corrections while retaining this prescribed sum.

Proof. We compute group homology on the original integral basis, replace its translation differential by a chain-isomorphic logarithmic differential, and read off the image in moment degree zero.

Step 1. We compute group homology using the Koszul resolution on the original integral period basis. The chosen basis identifies the group algebra with

$$\mathbb{Q}[\Delta] = \mathbb{Q}[t_1^{\pm 1}, \dots, t_g^{\pm 1}], \quad t_i = T_{\delta_i}.$$

The Koszul complex on $t_1 - 1, \dots, t_g - 1$ is a free resolution of the trivial module $\mathbb{Q}$. To see its exactness, quotienting by $t_1 - 1, \dots, t_b - 1$ leaves the Laurent polynomial ring in the remaining $g - b$ variables. Multiplication by $t_{b+1} - 1$ on this quotient is injective. Adding that generator to the Koszul complex forms the mapping cone of this multiplication. If the preceding complex has homology only in degree zero, the cone has first homology equal to that kernel, degree-zero homology equal to the quotient, and all higher homology zero. Induction from the empty sequence proves the claim, and the final quotient is $\mathbb{Q}$.

By Theorem 3.2, tensoring this free resolution with $M_p(R; U)$ gives a complex $K^{\exp}$ with

$$K_b^{\exp} = \bigwedge^b H \otimes E_p(R; U).$$

Put $D_i = D_{\delta_i}$ and $f_i = \exp(D_i) - \text{id}$. Its differential is

$$d_{\exp}(\delta_I \otimes z) = \sum_{c=1}^b (-1)^{c-1} \delta_{I \setminus \{i_c\}} \otimes f_{i_c} z, \quad I = \{i_1 < \dots < i_b\}.$$

This computes $H_*(\Delta; M_p(R; U))$. For the trivial module B , all the operators $t_i - 1$ act as zero, so the corresponding complex has zero differential and gives $H_a(\Delta; B) = \bigwedge^a H \otimes B$.

Step 2. We construct a finite chain isomorphism between the translation and logarithmic differentials. Put $\nu = m - p + 1$. The operators in the statement satisfy

$$f_i = D_i F_i, \quad F_i^{-1} = \sum_{b=0}^{\nu-1} (-1)^b (F_i - \text{id})^b.$$

Indeed, $D_i^\nu = 0$, and $F_i - \text{id}$ raises moment degree by at least one, so its ν -th power is zero. All the D_i , F_i , and their inverses commute.

Let K^D have the same chain groups as $K^{\exp}$, with differential

$$d_D(\delta_I \otimes z) = \sum_{c=1}^b (-1)^{c-1} \delta_{I \setminus \{i_c\}} \otimes D_{i_c} z.$$

Commutativity of the D_i gives $d_D^2 = 0$: the two orders of deleting any pair of indices have opposite signs. Define

$$\Phi_b : K_b^{\exp} \longrightarrow K_b^D, \quad \Phi_b(\delta_I \otimes z) = \delta_I \otimes \left(\prod_{i \in I} F_i \right) z.$$

For each term deleting i_c from I we have

$$D_{i_c} \prod_{i \in I} F_i = \left(\prod_{i \in I \setminus \{i_c\}} F_i \right) f_{i_c}.$$

The deletion signs in both differentials agree. This identity proves $d_D \Phi_b = \Phi_{b-1} d_{\exp}$. Thus Φ is a chain isomorphism, and its inverse multiplies the I -component by $\prod_{i \in I} F_i^{-1}$. Since F_i and F_i^{-1} have constant term id and all their other terms raise moment degree, both isomorphisms preserve projection to degree zero.

Step 3. We identify the coefficient-sum image with the eigenwave kernel and construct an explicit lift. Let z be a cycle in $K_a^{\exp}$ and decompose its image as

$$\Phi_a z = \sum_{k=0}^{m-p} z_k, \quad z_k \in \bigwedge^a H \otimes \bigwedge^k R \otimes \bigwedge^{p+k} U.$$

The differential d_D raises moment degree by one. The moment-degree one component of $d_D \Phi_a z = 0$ is therefore

$$d_D z_0 = W_{a,p}(z_0) = 0.$$

Projection to degree zero is unchanged by Φ_a , so z_0 is the coefficient sum of z . This proves that every homology class in the image belongs to $\ker W_{a,p}$. If $p = m$, the positive moment degrees and all D_i are zero, and this condition is the zero equation with zero target.

Conversely, regard $y \in \ker W_{a,p}$ as a chain of K_a^D concentrated in moment degree zero. Its full differential is $d_D y = W_{a,p}(y) = 0$. It is therefore a cycle, and

$$\Phi_a^{-1} y = \sum_{|I|=a} \delta_I \otimes \left(\prod_{i \in I} F_i^{-1} \right) \beta_I = L_a(y)$$

is a cycle in $K_a^{\exp}$ with coefficient sum y . The target B -complex has zero differential, so this equality is also the exact equality of the induced homology classes. In particular, the coefficient

sum of every boundary is zero. For $a = 0$ the same argument uses the empty product and the zero differential out of degree zero.

All corrections are finite polynomials in operators on the existing Δ -module. The free resolution is over $\mathbb{Q}[\Delta]$ throughout; $H = \Delta \otimes \mathbb{Q}$ only records its rational exterior basis. Thus the lift retains the original acting group. $\square$

For $1 \leq r \leq d$ in Set-up 2.1, take $U = V$, $R = S$, $p = r$, $\Delta = \Gamma$, and $a = r$. The inclusion of H into $R \otimes U$ is then the actual period inclusion j , and $W_{r,r}$ is exactly W_r from Definition 2.5. Theorem 3.3 therefore gives a finite lift of each $w \in \ker W_r$ in $H_r(\Gamma; M_r(S; V))$ with coefficient sum exactly w .

4. FINITE SPECIALIZATION AND SWEEPS

We retain the incidence-chain conventions of Definition 2.3. We construct two operations on finite chains: specialization of their real parameters by a rational linear map, and sweeping a parameter-dependent chain between two heights. Neither operation requires the specialized cells to remain convex with their original face structure. We first give the finite geometric arguments that replace their signed images by admissible polytopes.

For a rational vector space U , write $U_{\mathbb{R}} = U \otimes_{\mathbb{Q}} \mathbb{R}$. An oriented polytope P with constant coefficient β defines the integration functional $\omega \mapsto \beta \int_P \omega$ on smooth compactly supported forms of its dimension. We use these functionals only to compare finite chains.

Lemma 4.1. *Let U and B be finite-dimensional rational vector spaces, let $R \subset \mathbb{R}$ be finite-dimensional over $\mathbb{Q}$, and let $0 \leq k \leq \dim U$. Then the following hold.*

- (i) *Every finite list of rational-faced polytopes in $U_{\mathbb{R}}$ with vertices in $R \otimes U$ has a finite admissible common refinement.*
- (ii) *Integration is injective on finite oriented k -chains of these polytopes with coefficients in B , modulo orientation reversal, coefficient linearity, and finite admissible subdivision. It commutes with boundary.*

For every integer p with $k \leq p \leq \dim U$, the same conclusions hold on the incidence coefficient subspaces defining $C_{k,p}(R; U)$.

Thus an integration identity between admissible finite chains is an identity in their algebraic subdivision quotient.

Proof. Step 1. We first verify that the required cuts keep their vertices in the parameter space. Fix rational coordinates on U . For a rational subspace $L \subset U$, rational linear equations for L give

$$(4.1) \quad (R \otimes U) \cap L_{\mathbb{R}} = R \otimes L.$$

Indeed, solving those equations uses the same rational row operations on R -valued coordinate vectors as on rational coordinate vectors. If an admissible polytope has tangent L and a vertex z , rational coordinates on $z + L_{\mathbb{R}}$ therefore give R -valued coordinates for all its vertices. Every facet has a rational normal, and its offset is in R , as follows by evaluating that normal at a facet vertex.

Record rational affine equations for the carriers of all the given polytopes, and extend their facet equations rationally to $U_{\mathbb{R}}$. All offsets lie in R . Cut each polytope by this finite list of hyperplanes, omitting a cut when its hyperplane contains the entire carrier. The resulting pieces, together with their faces, form a common refinement. Each vertex is determined in its rational affine carrier by an invertible rational linear system with right-hand side in R . Its coordinates are therefore in R . The pieces are bounded convex polytopes, and every face direction is rational. This proves the first assertion without assuming that R is a field.

Step 2. We prove that integration detects finite chains and respects the incidence subspaces. Integration respects orientation, coefficient linearity, and subdivision. The formula

$$\int_P d\omega = \int_{\partial P} \omega$$

follows first on an affine simplex by integration by parts, and then on a polytope by any ordinary finite triangulation: the interior facets cancel with opposite orientations. This triangulation is used for integration only.

To prove injectivity, refine a finite presentation as above, choose one orientation on each resulting k -cell, and combine coincident coefficients. If a coefficient remains nonzero, choose a point in the relative interior of its cell away from the boundaries of the refined cells and from every distinct k -dimensional carrier. A smooth form supported in a small neighborhood of that point, with positive integral on the chosen orientation, detects this coefficient after pairing with a suitable element of $(B \otimes \mathbb{R})^*$. Thus a zero integration functional has zero coefficient on every refined cell. In degree zero the same argument is the independence of point masses.

Refinement preserves the tangent of every top-dimensional piece of a subdivided cell. Its incidence coefficient is consequently still allowed. On a facet of tangent $L' \subset L$, the inclusion

$$\bigwedge^k L \wedge \bigwedge^{p-k} U \subset \bigwedge^{k-1} L' \wedge \bigwedge^{p-k+1} U$$

follows by extending a basis of L' to one of L . Boundary thus retains an allowed coefficient, and the preceding arguments apply to the incidence subspaces. $\square$

Lemma 4.2. *Let U be a finite-dimensional rational vector space, let $R \subset \mathbb{R}$ be finite-dimensional over $\mathbb{Q}$, and let $\Pi = z + L_{\mathbb{R}}$ be an oriented affine space with $z \in R \otimes U$ and $L \subset U$ rational of dimension $k \geq 1$. Let $H_1, \dots, H_s$ be affine hyperplanes in Π with rational normals and R -valued offsets in rational coordinates based at z . Then the following hold.*

- (i) *A finite signed sum of ordinary affine k -simplices in Π whose boundary integration functional is supported on $\bigcup_i H_i$ has the same integration functional as a finite integer linear combination of the closures of bounded chambers of this hyperplane arrangement. These closures are rational-faced polytopes with vertices in $R \otimes U$.*
- (ii) *A compactly supported finite polyhedral k -chain in Π with coefficients in a finite-dimensional real vector space and zero boundary integration functional has zero integration functional.*

The first assertion supplies admissible images. The second gives uniqueness in their fixed affine carrier.

Proof. Refine the images of the ordinary simplices by all their facet hyperplanes. Relative to the orientation of Π , their sum has an integer-valued, piecewise constant multiplicity function with compact support. At a generic common facet of two small regions, integration by parts identifies the boundary coefficient with the jump of this multiplicity, up to the induced orientation sign.

Inside an open chamber of the H_i , this jump is zero. Two points of the chamber can be joined by a polygonal path transverse to the finitely many auxiliary walls and avoiding their intersections of codimension at least two. Such a path is obtained by choosing its intermediate points outside finitely many proper affine subspaces; in dimension one it is the interval between the points. Following the zero jumps shows that the multiplicity is constant on the chamber. An unbounded chamber contains points outside the compact support, so its multiplicity is zero. The closures of the remaining chambers give the asserted replacement, since the omitted walls have zero k -dimensional measure. Their vertices and face directions have the required properties by the rational linear-system argument in Lemma 4.1.

If the boundary is zero everywhere, the same argument without the H_i makes the multiplicity constant throughout Π . Compact support forces that constant to be zero. Apply linear functionals on the coefficient space to obtain the second assertion with vector coefficients. Ordinary triangulations reduce finite polyhedral chains to the simplex case. $\square$

Theorem 4.3. *Let U be a finite-dimensional rational vector space, let $R_0, R_1 \subset \mathbb{R}$ be finite-dimensional rational subspaces, let $f : R_0 \rightarrow R_1$ be rational-linear, and let $0 \leq p \leq \dim U$. There is a canonical chain map*

$$\mathrm{Sp}_f : C_{\bullet,p}(R_0; U) \longrightarrow C_{\bullet,p}(R_1; U)$$

which sends a point x to $(f \otimes \mathrm{id})x$, retains exterior coefficients, and sends every positive-dimensional cell into a finite signed chain in its specialized rational affine carrier. It commutes with every finite admissible subdivision, including folds and rank collapse. The maps satisfy $\mathrm{Sp}_{\mathrm{id}} = \mathrm{id}$, $\mathrm{Sp}_g \mathrm{Sp}_f = \mathrm{Sp}_{g \circ f}$ whenever defined, and

$$\mathrm{Sp}_f T_x = T_{(f \otimes \mathrm{id})x} \mathrm{Sp}_f.$$

If f fixes a subspace $R_ \subset R_0 \cap R_1$ pointwise, then it fixes every chain with vertices in $R_* \otimes U$.*

Specialization therefore preserves exact finite-chain identities even when the order of the real parameters changes. All exterior coefficients remain unchanged.

Proof. We construct the image of a cell in its fixed rational carrier, prove compatibility with boundary and subdivision, and then establish the composition and relative identities. Put $F = f \otimes \mathrm{id}_U$.

Step 1. We construct a finite signed image of each cell in its prescribed rational carrier. Let P be an oriented admissible k -polytope with rational tangent L . For a face Q with tangent L_Q and a vertex v_Q , (4.1) gives

$$F(v) - F(v_Q) \in R_1 \otimes L_Q \quad \text{for every vertex } v \text{ of } Q.$$

In particular, all specialized vertices of Q lie in $F(v_Q) + (L_Q)_{\mathbb{R}}$.

Triangulate the ordinary real polytope P by its face barycenters, taking the barycenter of each nonempty face to be the arithmetic average of its vertices. Send that barycenter to the average of the F -images of the same vertices. Extend affinely over each simplex of the barycentric triangulation. These affine maps agree on common faces, and the image of every face Q lies in its specified specialized carrier. The auxiliary simplices need not have rational directions.

Suppose $k \geq 1$. All top images lie in the oriented k -space

$$\Pi_P = F(v_P) + L_{\mathbb{R}},$$

where v_P is any vertex of P and the orientation is transported from L . Sum the full-rank image simplices with their orientation signs. A rank-deficient image contributes zero in degree k . For an affine simplex map ϕ , pullback commutes with differentiation, so simplex integration by parts gives

$$\partial(\phi_{\#}[\sigma]) = \phi_{\#}(\partial[\sigma])$$

as integration functionals, also when ϕ has rank less than k . Thus a collapsed top simplex has zero total boundary image; discarding its top image leaves no uncanceled boundary term.

The interior faces of the domain triangulation cancel. The remaining boundary is supported on the specialized carriers of the original facets of P . These are finitely many hyperplanes in Π_P . Their normals are rational, and their offsets in rational coordinates based at $F(v_P)$ are in R_1 : evaluate a normal at the image of one original facet vertex. Lemma 4.2 therefore replaces the signed image by

$$\mathrm{Sp}_f(P, \beta) = \sum_D n_D [D, \beta],$$

where D runs over the closures of the bounded chambers with nonzero multiplicity, $n_D \in \mathbb{Z}$, and β is the original coefficient. Every D has tangent L , rational face directions, and vertices in $R_1 \otimes U$. Thus its retained coefficient is incident. If the mapped original vertices span a space of dimension less than k , all auxiliary images have rank less than k , and this chain is zero. For $k = 0$, define $\mathrm{Sp}_f([x, \beta]) = [F(x), \beta]$.

Step 2. We prove that specialization commutes with boundary and respects admissible subdivisions. The barycentric triangulation induced on an original facet is its own face-barycentric

triangulation. The preceding boundary identity therefore identifies the boundary functional of $\text{Sp}_f(P, \beta)$ with that of $\text{Sp}_f(\partial(P, \beta))$. Both sides are finite admissible chains with the full retained coefficient. Lemma 4.1 gives their equality in the incidence quotient.

We prove compatibility with every actual admissible subdivision by induction on k . The assertion holds for points. Let $P = \sum_j P_j$ be such a subdivision in positive degree, with the induced orientations and coefficient β . The boundary identity

$$\partial(P, \beta) = \sum_j \partial(P_j, \beta)$$

holds after a finite common refinement of its facets: each interior facet has two opposite incidences and the exterior facets subdivide ∂P . By the induction hypothesis and the generator boundary identity, the chain

$$\text{Sp}_f(P, \beta) - \sum_j \text{Sp}_f(P_j, \beta)$$

has zero boundary. Every vertex of every P_j differs from v_P by an element of $R_0 \otimes L$. Hence all its terms lie in the same k -dimensional carrier Π_P . They have compact support. Lemma 4.2(ii), followed by Lemma 4.1, makes their difference zero as a finite admissible chain.

Orientation reversal and coefficient linearity are respected by the construction. It consequently descends to the asserted chain map. This argument also explains why new subdivision vertices cause no difficulty: the two auxiliary maps need not agree inside P ; their boundary agreement and uniqueness in the carrier suffice.

Step 3. We establish independence of auxiliary choices, composition, and the relative specialization identities. The same induction proves independence from any alternative finite face-compatible auxiliary triangulations or affine realizations using the prescribed point images and face carriers. In positive degree, two such images have equal boundaries by induction, and their difference is a compact top-dimensional cycle in Π_P . The two lemmas again make it zero.

For $f = \text{id}$, all barycenters are fixed and the auxiliary map is the identity on each simplex. Thus $\text{Sp}_{\text{id}} = \text{id}$. To prove composition, start with its equality on points and argue by degree. For a k -cell P , the two chains $\text{Sp}_g \text{Sp}_f(P, \beta)$ and $\text{Sp}_{g \circ f}(P, \beta)$ have equal boundaries by the lower-degree identity. Their carriers agree. Indeed, a vertex y of a rectified intermediate cell satisfies

$$y - F(v_P) \in R_1 \otimes L,$$

so applying $g \otimes \text{id}$ puts every final term in $((g \circ f) \otimes \text{id})(v_P) + L_{\mathbb{R}}$. The difference is compact and has zero boundary in this k -space, so it vanishes. This also covers an intermediate collapse, when one of the chains is zero.

Translation by x adds $F(x)$ to every mapped barycenter and to every specialized facet carrier. It translates the entire chamber construction by $F(x)$, proving the translation identity. If all vertices lie in $R_* \otimes U$ and f fixes R_* , every mapped barycenter is unchanged and the auxiliary map is the identity. This proves the fixed-chain assertion. When $U = 0$, only the point construction is needed. Every construction and refinement used above is finite. $\square$

We now distinguish a height direction from the horizontal space. The coefficient of a cap must record the velocity of its affine carrier as well as the horizontal coefficient.

Definition 4.4. Let $U = \mathbb{Q}v_0 \oplus W$, let $\alpha(v_0) = 1$ and $\alpha(W) = 0$, let $R \subset \mathbb{R}$ be finite-dimensional over $\mathbb{Q}$, and let $\tau \in \mathbb{R}$ be independent of R over $\mathbb{Q}$. Set $T = R \oplus \mathbb{Q}\tau$. Write each vertex x of a cell over T uniquely as $a_x + \tau b_x$, with $a_x \in R \otimes W$ and $b_x \in W$. For $t \in R$ let $e_t : T \rightarrow R$ fix R and send τ to t . Fix integers $0 \leq k \leq p \leq \dim W$. For a k -cell P of direction $L \subset W$, coefficient $\beta \in \bigwedge^k L \wedge \bigwedge^{p-k} W$, and a vertex x_0 of P , the *full cap* $\mathfrak{m}_t(P, \beta)$ is $\text{Sp}_{e_t}(P)$ translated by tv_0 , carrying coefficient $(v_0 + b_{x_0}) \wedge \beta$.

The cap lies in the height hyperplane $\alpha = t$ and has coefficient degree $p + 1$. We write ι_α for contraction on exterior coefficients, characterized by $\iota_\alpha u = \alpha(u)$ and the graded Leibniz rule. It does not change a cell's support or orientation.

Theorem 4.5. *In the setting of Definition 4.4, let $0 \leq p \leq \dim W$. The full cap is independent of its anchor and defines a linear map on $C_{k,p}(T; W)$. For $a, b \in R$ there are linear maps*

$$\text{Sw}_a^b : C_{k,p}(T; W) \longrightarrow C_{k+1,p+1}(R; U) \quad (0 \leq k \leq p)$$

whose output is finite and whose nonzero cells over (P, β) have rational tangent $L + \mathbb{Q}(v_0 + b_{x_0})$ and the lifted exterior coefficient $(v_0 + b_{x_0}) \wedge \beta$ multiplied by a signed integer. They satisfy

$$\begin{aligned} \partial \mathbb{M}_t c &= \mathbb{M}_t \partial c, & \iota_\alpha \mathbb{M}_t c &= T_{tv_0} \text{Sp}_{e_t} c, \\ \partial \text{Sw}_a^b c &= \mathbb{M}_b c - \mathbb{M}_a c - \text{Sw}_a^b \partial c, & \text{Sw}_a^e &= \text{Sw}_a^b + \text{Sw}_b^e \quad (e \in R). \end{aligned}$$

These identities respect all finite admissible subdivisions and hold through folds and collapse. If $g : R \rightarrow R'$ is rational-linear and $\tilde{g} : R \oplus \mathbb{Q}\tau \rightarrow R' \oplus \mathbb{Q}\tau'$ extends g by $\tau \mapsto \tau'$, with τ' independent of R' , then

$$\text{Sp}_g \mathbb{M}_t = \mathbb{M}_{g(t)} \text{Sp}_{\tilde{g}}, \quad \text{Sp}_g \text{Sw}_a^b = \text{Sw}_{g(a)}^{g(b)} \text{Sp}_{\tilde{g}}.$$

For finitely many such families on consecutive intervals, the boundary of the sum of their sweeps retains, at every junction, the incoming full cap minus the outgoing full cap. If the specialized horizontal chains at that junction agree, the cap difference has zero α -contraction and is an incident $(p + 1)$ -vector-valued chain in W after translation of its height to zero.

The sweep formula keeps the entire exterior coefficient on both caps and every side face. In particular, matching horizontal slices at a junction can leave a nonzero horizontal cap difference.

Proof. We first determine the rational carriers and their coefficients. We then rectify a finite prism to construct the sweep, prove its identities by uniqueness in the swept carrier, and identify the residual at a junction.

Step 1. We determine the rational swept carriers and their full lifted coefficients. Suppose x, y are vertices of a face with rational tangent L_Q . Applying (4.1) over T gives $x - y \in T \otimes L_Q$. Apply rational annihilators of L_Q to the equality

$$x - y = (a_x - a_y) + \tau(b_x - b_y).$$

The first term is R -valued and the second has rational τ -coefficient. Independence of τ over R gives

$$(4.2) \quad a_x - a_y \in R \otimes L_Q, \quad b_x - b_y \in L_Q.$$

The same argument applies to every vertex introduced by an admissible subdivision.

For a k -cell P of tangent L , choose a vertex x_0 and put

$$w_P = v_0 + b_{x_0}, \quad \eta_P = w_P \wedge \beta.$$

Since $\ell \wedge \beta = 0$ for every $\ell \in L$, equation (4.2) proves

$$(4.3) \quad (v_0 + b_x) \wedge \beta = \eta_P, \quad \iota_\alpha \eta_P = \beta.$$

The second identity uses $\alpha(w_P) = 1$ and $\iota_\alpha \beta = 0$. Thus η_P is independent of the anchor. It contains the full $(k + 1)$ -dimensional rational direction $L + \mathbb{Q}w_P$. For a facet of P , its anchor can be chosen among the vertices of P , and its lifted coefficient with the retained β is again η_P . The coefficient β is incident to that facet by the exterior-power inclusion used in Lemma 4.1.

The lift also recovers a given full incident coefficient. For

$$\eta \in \bigwedge_{k+1} (L + \mathbb{Q}w_P) \wedge \bigwedge_{p-k} U,$$

one has $w_P \wedge \eta = 0$. Contracting this equality gives

$$\eta = w_P \wedge \iota_\alpha \eta.$$

In the decomposition $U = \mathbb{Q}w_P \oplus W$, extending a basis of L to one of W shows that

$$\iota_\alpha \eta \in \bigwedge^k L \wedge \bigwedge^{p-k} W.$$

This explains why the lift retains the original coefficient of a swept cell, rather than only its contraction.

Step 2. We show that the full caps commute with boundary and descend to the subdivision quotient. Theorem 4.3 sends P to a finite signed chain in its specialized L -carrier. After translation by tv_0 , η_P is an allowed coefficient on every surviving cell. On each original facet it is the same coefficient, by (4.3). The boundary identity for specialization therefore gives

$$\partial \mathbb{M}_t(P, \beta) = \mathbb{M}_t \partial(P, \beta).$$

If P is subdivided, the anchors of all its top pieces differ from x_0 by elements of $T \otimes L$. Their lifted coefficients are again η_P . Subdivision compatibility for specialization now proves compatibility for caps. Orientation and coefficient linearity are preserved as well. This defines

$$\mathbb{M}_t : C_{k,p}(T; W) \longrightarrow C_{k,p+1}(R; U),$$

and contraction gives the displayed identity of the theorem by (4.3).

Step 3. We rectify a finite prism to construct the sweep and compute its boundary. First let $a < b$. Use the face-barycentric triangulation of P as an ordinary real domain. Form $[a, b] \times P$ with height first orientation. A global ordering of the barycentric vertices gives a staircase triangulation of each simplex times the interval; these triangulations agree on common faces. At the two endpoint levels send an original vertex x to

$$tv_0 + a_x + tb_x \quad (t = a, b),$$

and use the corresponding vertex averages for face barycenters. Extend affinely on every prism simplex. This is a finite piecewise affine domain map.

All image simplices lie in the fixed rational affine carrier

$$(4.4) \quad \Pi_P = a_{x_0} + L_{\mathbb{R}} + \mathbb{R}w_P.$$

Indeed, subtracting the anchor track from the image of a vertex leaves $(a_x - a_{x_0}) + t(b_x - b_{x_0}) \in L_{\mathbb{R}}$. The carrier has dimension $k + 1$ because $\alpha(w_P) = 1$ and $\alpha(L) = 0$. Orient it by w_P followed by the orientation of L ; this orientation is independent of the anchor.

The cap images lie in the two hyperplanes $\alpha = a, b$ of Π_P . For each original facet Q of tangent L_Q , all its side images lie in

$$a_{x_Q} + (L_Q)_{\mathbb{R}} + \mathbb{R}(v_0 + b_{x_Q}).$$

This is a rational affine k -plane in Π_P , hence a hyperplane. Its anchor is R -valued. Thus the cap and original side carriers form a finite list of hyperplanes with rational normals and R -valued offsets.

Push forward each auxiliary simplex as an integration functional. Full-rank simplices contribute their signed images; lower-rank simplices have zero top functional. As in the specialization proof, affine pullback and simplex integration by parts show that the total boundary image of each collapsed simplex is zero. Interior faces cancel, leaving the boundary supported on the cap and side hyperplanes just listed. Lemma 4.2(i) replaces this signed top image by a finite integer combination of bounded chamber closures. Equip each chamber with η_P and the orientation of Π_P . We define the resulting admissible chain to be $\text{Sw}_a^b(P, \beta)$. Its vertices belong to $R \otimes U$, and its tangent and coefficient have exactly the asserted form.

The oriented product boundary is

$$\partial([a, b] \times P) = ([b] - [a]) \times P - [a, b] \times \partial P.$$

The endpoint maps are the barycentric specialization maps, so their rectifications give $\mathbb{M}_b(P, \beta)$ and $\mathbb{M}_a(P, \beta)$. On an original facet the restricted prism construction uses that facet's barycenters and has the same full coefficient η_P . Induction on k therefore identifies the rectified side with the

sweep of the oriented facet. For $k = 0$, the construction is the segment from $av_0 + a_x + ab_x$ to $bv_0 + a_x + bb_x$, with its retained coefficient, and there are no side terms. This starts the induction. The product formula gives the equality of integration functionals, and Lemma 4.1 gives the exact finite-chain identity

$$(4.5) \quad \partial \operatorname{Sw}_a^b(P, \beta) = \mathfrak{m}_b(P, \beta) - \mathfrak{m}_a(P, \beta) - \operatorname{Sw}_a^b \partial(P, \beta).$$

Set $\operatorname{Sw}_a^a = 0$ and, for reversed endpoints, set $\operatorname{Sw}_a^b = -\operatorname{Sw}_b^a$. The same formula follows with the oppositely oriented interval. If $W = 0$, then $p = k = 0$, the horizontal point is the origin, its cap is tv_0 with coefficient $v_0\beta$, and the sweep is the oriented segment between the two heights. Thus this case is included directly.

Step 4. We justify the independence of prism choices and all source subdivisions simultaneously by induction on k . For points there is only the specified oriented track. Suppose the claims hold in lower degree. Two compatible prism triangulations for P have the same caps, and their side chains agree by induction. Equation (4.5) makes their difference a compact cycle in the $(k+1)$ -space (4.4). Lemma 4.2(ii) and faithful integration make this difference zero.

Now subdivide $P = \sum_j P_j$ admissibly. Every top piece has tangent L and, by (4.2), the same swept carrier and lifted coefficient. In

$$\operatorname{Sw}_a^b(P, \beta) - \sum_j \operatorname{Sw}_a^b(P_j, \beta),$$

the cap terms cancel by cap compatibility. The side terms cancel by the lower-degree subdivision identity and the opposite incidences of internal subdivision facets. Compact top-dimensional uniqueness again makes the difference zero. The construction respects orientation and is linear in β , since the chamber multiplicities depend only on the cell geometry. The sweep is therefore a linear operator on the exact incidence quotient.

For interval additivity, point tracks satisfy oriented segment additivity for any order of a, b, e : subdivide at the three heights and cancel opposite occurrences. Inductively the difference

$$\operatorname{Sw}_a^e(P, \beta) - \operatorname{Sw}_a^b(P, \beta) - \operatorname{Sw}_b^e(P, \beta)$$

has zero boundary. Its cap terms cancel directly, and its side terms cancel by the lower-degree additivity identity. All terms have the same carrier (4.4), which is independent of the endpoints. The uniqueness argument proves additivity. None of these arguments imposes a rank condition on the intermediate specialized horizontal cells.

Step 5. We verify the naturality of caps and sweeps under changes of parameters, including their full coefficients. For the two evaluation maps, one has

$$g(e_t(s + q\tau)) = g(s) + qg(t) = e_{g(t)}(\tilde{g}(s + q\tau)) \quad (s \in R, q \in \mathbb{Q}).$$

Functoriality of specialization therefore identifies the underlying horizontal chains in the two cap expressions. Translation by tv_0 specializes to translation by $g(t)v_0$.

We must also compare their full coefficients after rectification. For one original cell with anchor x_0 , every vertex y of a nonzero cell in $\operatorname{Sp}_{\tilde{g}}(P)$ satisfies

$$y - ((g \otimes \operatorname{id}_W)a_{x_0} + \tau'b_{x_0}) \in (R' \oplus \mathbb{Q}\tau') \otimes L.$$

Write $y = a'_y + \tau'b'_y$. Independence of τ' gives $b'_y - b_{x_0} \in L$. Its lifted coefficient is consequently

$$(v_0 + b'_y) \wedge \beta = (v_0 + b_{x_0}) \wedge \beta = \eta_P.$$

With the signed multiplicities retained, this proves cap naturality, including all coincident and collapsed terms.

For sweeps, start with points. The two expressions are oriented tracks with the same specialized endpoints and full coefficient. Theorem 4.3 identifies them even if the endpoints coincide or their

order reverses. In higher degree, (4.5), cap naturality, and the induction hypothesis on the side terms give equal boundaries. Both chains lie in

$$(g \otimes \text{id}_W)a_{x_0} + L_{\mathbb{R}} + \mathbb{R}w_P.$$

On the right, the preceding calculation of a'_y and b'_y places every rectified source piece in this same swept carrier with the same lifted coefficient. On the left, specialization preserves that rational carrier direction. Their difference is a compact top-dimensional cycle and hence zero. If the intermediate source cell collapses, the right-hand side is zero and the same argument still applies. This proves both naturality identities without any order or injectivity assumption on g .

Step 6. We identify the full cap residuals at interval junctions and show when they are horizontal. Let $t_0 < \dots < t_m$ be in R and let c_i be the source chain for the interval $[t_{i-1}, t_i]$. Each interval may use its own independent parameter; denote its operators by $\mathbb{M}_t^{(i)}$ and $\text{Sw}^{(i)}[a, b]$, with the latter denoting the sweep from a to b . For $1 \leq i < m$, set

$$K_i = \mathbb{M}_{t_i}^{(i)} c_i - \mathbb{M}_{t_i}^{(i+1)} c_{i+1}.$$

Summing (4.5) gives

$$\begin{aligned} \partial \sum_{i=1}^m \text{Sw}^{(i)}[t_{i-1}, t_i] c_i &= \mathbb{M}_{t_m}^{(m)} c_m - \mathbb{M}_{t_0}^{(1)} c_1 + \sum_{i=1}^{m-1} K_i \\ &\quad - \sum_{i=1}^m \text{Sw}^{(i)}[t_{i-1}, t_i] \partial c_i. \end{aligned}$$

Thus every incoming full cap minus its outgoing full cap occurs with the stated sign.

If the specialized horizontal chains agree at t_i , contraction of K_i is zero by (4.3). Translate its height to zero and take a finite common admissible refinement in W . On a resulting k -cell of tangent L , write its combined coefficient uniquely as

$$\eta = \eta_0 + v_0 \wedge \eta_1, \quad \eta_0 \in \bigwedge^{p+1} W, \quad \eta_1 \in \bigwedge^p W.$$

Contraction gives η_1 . Faithful integration, or coefficient detection on this refinement, therefore gives $\eta_1 = 0$ on every cell. Finally, extend a basis of L to a basis of W and then add v_0 . In that basis,

$$\left(\bigwedge^k L \wedge \bigwedge^{p+1-k} U \right) \cap \bigwedge^{p+1} W = \bigwedge^k L \wedge \bigwedge^{p+1-k} W.$$

The remaining coefficient is horizontal and still incident to its cell. All refinements, chamber sums, and prism constructions are finite, and all final vertices belong to the original space $R \otimes U$. $\square$

Remark 4.6. The residual in Theorem 4.5 can be nonzero. Take $W = \mathbb{Q}e_1 \oplus \mathbb{Q}e_2$, $R = \mathbb{Q}$, and the horizontal point with coefficient $\beta = e_1$. Move the point as te_1 on $[-1, 0]$ and as te_2 on $[0, 1]$. At height zero the two horizontal chains agree. Their full cap coefficients are $v_0 \wedge e_1$ and $(v_0 + e_2) \wedge e_1$, respectively. The incoming cap minus the outgoing cap has coefficient $-e_2 \wedge e_1$, which is nonzero and has zero α -contraction.

5. FINITE LOCAL FILLINGS

We retain the incidence complex of Definition 2.3 and the specialization and sweep operations of the preceding section. We prove that every closed chain of positive degree below its exterior coefficient degree has a finite filling. The degree-one case uses all the endpoint moments. Higher degrees then follow by slicing, with a second induction to fill the horizontal differences between full caps.

We first record the parameter elimination used to terminate both constructions.

Lemma 5.1. *Let W be a finite-dimensional rational vector space, let $R \subset \mathbb{R}$ be a nonzero finite-dimensional rational subspace, and let $1 \leq k \leq s \leq \dim W$. Let $c \in C_{k,s}(R; W)$ satisfy $\partial c = 0$, and let $h \in R$. For any $\tau \in \mathbb{R}$ independent of R over $\mathbb{Q}$ and any $t \in R$, denote by $e_t^\tau : R \oplus \mathbb{Q}\tau \rightarrow R$ the map fixing R and sending τ to t . Then there exist an integer $\ell \geq 1$, heights $h = h_0 < h_1 < \dots < h_\ell$ in R , parameters $\tau_i \in (h_{i-1}, h_i)$ independent of R , and finite closed chains*

$$c_i \in C_{k,s}(R \oplus \mathbb{Q}\tau_i; W) \quad (1 \leq i \leq \ell)$$

such that

$$\begin{aligned} \mathrm{Sp}_{e_{h_0}^{\tau_1}} c_1 &= c, \\ \mathrm{Sp}_{e_{h_i}^{\tau_i}} c_i &= \mathrm{Sp}_{e_{h_i}^{\tau_{i+1}}} c_{i+1} \quad (1 \leq i < \ell), \\ \mathrm{Sp}_{e_{h_\ell}^{\tau_\ell}} c_\ell &= 0. \end{aligned}$$

Thus a closed positive-dimensional horizontal chain can be brought to zero in finitely many intervals whose endpoint heights belong to R . The conclusion matches the specialized chains at each junction; the full cap differences will still be retained.

Proof. Choose a rational basis $r_1, \dots, r_\ell$ of R and change its individual signs so that $r_i > 0$. Set

$$h_i = h + \sum_{a=1}^i r_a.$$

Each open interval (h_{i-1}, h_i) contains a real number τ_i independent of R , since R is countable. Define a rational-linear map $f_i : R \rightarrow R \oplus \mathbb{Q}\tau_i$ by

$$f_i(r_a) = \begin{cases} 0, & a < i, \\ r_i + h_{i-1} - \tau_i, & a = i, \\ r_a, & a > i. \end{cases}$$

Let $c_i = \mathrm{Sp}_{f_i} c$. Theorem 4.3 makes this a finite closed chain. If $\pi_i : R \rightarrow R$ kills $r_1, \dots, r_i$ and fixes the remaining basis elements, with $\pi_0 = \mathrm{id}$, then

$$e_{h_{i-1}}^{\tau_i} f_i = \pi_{i-1}, \quad e_{h_i}^{\tau_i} f_i = \pi_i.$$

Functoriality proves the initial and junction identities. At the final endpoint $\pi_\ell = 0$. Every vertex of every original cell is then sent to the origin. Its barycentric image in the specialization construction is constant, so its positive-dimensional chain is zero. This proves the last identity.

For use with sweeps, the vertices of each c_i have unique expressions $a + \tau_i b$ with $a \in R \otimes W$ and $b \in W$. If two such vertices lie in a face of rational direction L , apply a rational covector annihilating L to their difference. Independence of τ_i from R shows separately that the difference of their a terms lies in $R \otimes L$ and the difference of their b terms lies in L . Thus all face-carrier equations persist under evaluation. After adjoining a rational height direction v_0 , the full sweeps of Theorem 4.5 apply, including when evaluation folds or collapses a cell. They have no side term because $\partial c_i = 0$, and all their vertices lie in $R \otimes (\mathbb{Q}v_0 \oplus W)$. $\square$

Theorem 5.2. *Let U be a finite-dimensional rational vector space, let $R \subset \mathbb{R}$ be finite-dimensional over $\mathbb{Q}$, and let $2 \leq p \leq \dim U$. Every finite closed chain in $C_{1,p}(R; U)$ is the boundary of a finite chain in $C_{2,p}(R; U)$. The filling retains rational coefficients, rational face directions, and vertices in the original space $R \otimes U$.*

The endpoint theorem fills each slice once all its moments vanish. Sweeping those fillings leaves horizontal cap differences, which are filled in a rational space of smaller dimension.

Proof. For each fixed $p \geq 2$, we induct on $n = \dim U$. We first show that every slice has all the required zero moments, then match and sweep its segment fillings, and finally fill the horizontal

residuals by the dimension induction. We include $n < p$ in the induction, when $\bigwedge^p U = 0$. If $R = 0$, there are no positive-dimensional cells. These are the initial cases.

Step 1. We prove that each slice has zero complete moment vector and hence admits a finite segment filling. Let $z \in C_{1,p}(R; U)$ be finite and closed. Choose a rational functional $\alpha \in U^*$ nonzero on every edge direction in a finite presentation of z . Such a choice avoids finitely many proper rational hyperplanes. Choose $v_0 \in U$ with $\alpha(v_0) = 1$ and set $W = \ker \alpha$, so $U = \mathbb{Q}v_0 \oplus W$. Orient each edge upward in height α , absorbing any orientation change into its coefficient. Write w_e for its rational direction normalized by $\alpha(w_e) = 1$, and β_e for its coefficient. Incidence gives

$$(5.1) \quad \gamma_e = \iota_\alpha \beta_e \in \bigwedge^{p-1} W, \quad \beta_e = w_e \wedge \gamma_e.$$

Indeed $w_e \wedge \beta_e = 0$, and contraction of this identity yields $\beta_e = w_e \wedge \iota_\alpha \beta_e$.

Let $t_0 < \dots < t_m$ be the distinct vertex heights. They belong to R . On an interval $I_j = (t_j, t_{j+1})$, the point at height t on each crossing edge has the form

$$x_e(t) = tv_0 + y_e(t), \quad y_e(t) = a_e + tb_e, \quad a_e \in R \otimes W, \quad b_e = w_e - v_0 \in W.$$

This follows by solving the edge-height equation with the rational vector w_e . Define the horizontal point chain

$$\zeta_j(t) = \sum_{e \text{ crossing } I_j} [y_e(t)] \otimes \gamma_e.$$

Choose $\tau_j \in I_j$ independent of R , and put $R_j = R \oplus \mathbb{Q}\tau_j$. Truncate z below height τ_j . The result is a finite admissible segment chain over R_j , whose boundary is

$$\sum_{e \text{ crossing } I_j} [x_e(\tau_j)] \otimes \beta_e.$$

All original endpoint terms cancel by $\partial z = 0$. By Theorem 3.2, the displayed boundary has zero moment in $E_p(R_j; U)$. Translate it by $-\tau_j v_0$. The moment remains zero because translation acts by the invertible operator $\exp(-D_{\tau_j v_0})$.

All translated positions are horizontal. In moment degree a , contraction of the U -exterior factor consequently gives

$$(\text{id} \wedge^a_{R_j} \otimes \iota_\alpha) \left(\sum_e \frac{1}{a!} D_{y_e(\tau_j)}^a \beta_e \right) = (-1)^a \sum_e \frac{1}{a!} D_{y_e(\tau_j)}^a \gamma_e = 0.$$

On the right the operators are those for the horizontal space W . The sign occurs because contraction passes the a horizontal position factors. The degrees available on the left are exactly

$$0 \leq a \leq n - p = \dim W - (p - 1).$$

They therefore exhaust $E_{p-1}(R_j; W)$. The full moment of $\zeta_j(\tau_j)$ vanishes. Applying Theorem 3.2 in W with coefficient degree $p - 1$ gives a finite chain

$$\widetilde{K}_j \in C_{1,p-1}(R_j; W), \quad \partial \widetilde{K}_j = \zeta_j(\tau_j).$$

Step 2. We match the slice fillings at consecutive endpoints and terminate the resulting family. Write $e_t^j : R_j \rightarrow R$ for evaluation $\tau_j \mapsto t$ fixing R , for $t \in R$. Every vertex of $\widetilde{K}_j$ is uniquely $a + \tau_j b$, with $a \in R \otimes W$ and $b \in W$. Its specializations form a signed affine family with rational velocities. Theorem 4.3 gives

$$\partial \text{Sp}_{e_t^j} \widetilde{K}_j = \zeta_j(t).$$

Here $\zeta_j(t)$ uses the displayed affine point formulas, including at the interval endpoints. In degree zero, equality at the independent parameter means equality of both the a and b coordinates of every canceling trajectory. Thus the boundary identity also holds as a formal identity of trajectories, including new coincidences and collapsed segments.

At a critical height the incoming and outgoing point chains agree. For the edges ending or beginning at one vertex, this is the α -contraction of its closedness equation in z ; edges passing

through that height contribute equally on both sides. The extreme point chains are zero by the same equation. We may consequently construct the fillings in height order. Start with $K_* = 0$ at t_0 . At the interval I_j , replace $\widetilde{K}_j$ by

$$(5.2) \quad K_j = \widetilde{K}_j + K_* - \text{Sp}_{e_{t_j}} \widetilde{K}_j,$$

regarding the last two terms as constant chains over R_j . Their difference is closed. The lower specialization of K_j is K_* and its boundary family is still $\zeta_j(t)$. Use its upper specialization as the next prescribed K_* . These identities match actual segment chains, with their full horizontal coefficients, in the finite subdivision quotient.

After the last interval, K_* is a closed chain in $C_{1,p-1}(R; W)$. Apply Lemma 5.1 at height t_m to continue it through finitely many closed families ending at zero. All new interval endpoints belong to R .

Step 3. We sweep the matched fillings with their full coefficients to recover the input edges. Apply Theorem 4.5 to every matched interval family and every terminal family, using horizontal coefficient degree $p - 1$. For one segment occurrence with endpoints $a(t) = a_0 + ta_1$ and $b(t) = b_0 + tb_1$, write

$$b(t) - a(t) = \lambda(t)u, \quad \lambda(t) = \lambda_0 + t\lambda_1, \quad \lambda_0 \in R, \quad \lambda_1 \in \mathbb{Q},$$

where $u \in W$ is a nonzero rational direction. Its coefficient δ belongs to $u \wedge \bigwedge^{p-2} W$. The swept coefficient is

$$(v_0 + a_1) \wedge \delta = (v_0 + b_1) \wedge \delta.$$

The equality follows from $b_1 - a_1 = \lambda_1 u$. This coefficient contains the two rational tangent directions of the swept polygon. In coordinates (t, r) that polygon is the signed region between $r = 0$ and $r = \lambda(t)$, mapped by

$$(t, r) \mapsto tv_0 + a(t) + ru.$$

For negative $\lambda(t)$ we take the negative of the region between $\lambda(t)$ and zero, with height first orientation. We split at any zero of a nonconstant λ ; that zero belongs to R since λ_1 is rational. This gives finitely many triangles or trapezoids with $R \otimes U$ vertices. An identically collapsed segment contributes zero. This describes concretely the signed sweep and its boundary formula, also when a segment reverses direction.

For an original interval, the side sweep of $\partial K_j = \zeta_j(\tau_j)$ is precisely the portion of z in that interval. Its point tracks have tangent $w_e = v_0 + b_e$, and (5.1) recovers their entire coefficient β_e . Terminal families are closed, so their side sweeps are zero. Let $\mathcal{D} \in C_{2,p}(R; U)$ be the sum of all sweeps. Their boundary identities give

$$(5.3) \quad \partial \mathcal{D} = \rho - z.$$

Here ρ is the sum of the full upper cap minus the full lower cap on every interval. In particular it contains the incoming full cap minus the outgoing full cap at each internal junction, and it includes both extreme caps.

Step 4. We fill the remaining horizontal residual by induction on the ambient dimension. By the contraction identity of Theorem 4.5, contraction of a full cap is the corresponding horizontal chain at that height. The endpoint matching and the zero extreme chains imply $\iota_\alpha \rho = 0$ on every height level. We use only this contraction identity: the uncontracted adjacent caps need not agree.

Take a finite common admissible subdivision on each of these levels. Such subdivisions keep R -valued vertices: carrier and facet equations have rational linear parts and constants in R , and the new vertices solve rational linear systems with those constants. On a horizontal edge of direction $\mathbb{Q}u$, the remaining coefficient η belongs to $u \wedge \bigwedge^{p-1} U$ and satisfies $\iota_\alpha \eta = 0$. Write

a tensor whose wedge with u is η as $\theta_0 + v_0 \wedge \theta_1$, with both θ_i horizontal. Contraction gives $u \wedge \theta_1 = 0$. Therefore

$$\eta \in u \wedge \bigwedge^{p-1} W.$$

Each level of ρ , translated into W , is consequently a chain in $C_{1,p}(R; W)$.

Equation (5.3) and $\partial z = 0$ give $\partial \rho = 0$. Distinct height hyperplanes are disjoint, so every level is separately closed. Translation by the negative of its height times v_0 preserves R -valued vertices. The induction in ambient dimension fills this translated level in W , with the same coefficient degree p ; if $\dim W < p$, its coefficient space is already zero. Restore the heights and sum these finite fillings to obtain $\mathcal{F}$ with $\partial \mathcal{F} = \rho$. Then

$$\partial(\mathcal{F} - \mathcal{D}) = z.$$

The slice parameters occur only in the descriptions of the families. Every actual sweep has endpoint heights in R , and Theorem 4.5 places all its vertices in $R \otimes U$. There are finitely many original intervals, parameter eliminations, and residual levels. The ambient induction decreases the dimension. All coefficients and directions remain rational throughout, which proves the asserted finite filling in the original parameter space. $\square$

Theorem 5.3. *Let U be a finite-dimensional rational vector space, let $R \subset \mathbb{R}$ be finite-dimensional over $\mathbb{Q}$, and let $1 \leq p \leq \dim U$. For every $0 < q < p$, every finite closed chain in $C_{q,p}(R; U)$ is the boundary of a finite chain in $C_{q+1,p}(R; U)$. In particular,*

$$H_q(C_{\bullet,p}(R; U)) = 0 \quad (0 < q < p).$$

The filling retains rational coefficients, rational face directions, and vertices in the original space $R \otimes U$.

The range $q < p$ allows a slice to be filled with one less exterior coefficient. The horizontal cap residual keeps coefficient degree p and is handled by decreasing the ambient dimension.

Proof. We induct first on q , proving the assertion for all coefficient degrees $p > q$, all finite rational ambient spaces, and all finite parameter spaces. The case $q = 1$ is Theorem 5.2. Fix $q \geq 2$ and assume the assertion in all smaller positive chain degrees. For each fixed $p > q$, we also induct on $n = \dim U$. As before, the coefficient spaces vanish for $n < p$, and the positive-degree chain spaces vanish for $R = 0$. We fill the slices by the induction on q , sweep matched families, and fill their horizontal residuals by the induction on n .

Step 1. We construct closed slices in one lower chain degree and fill them by the chain-degree induction. Let $z \in C_{q,p}(R; U)$ be finite and closed. Choose a rational height functional α nonzero on every edge direction of a finite polyhedral presentation. Choose $v_0 \in U$ with $\alpha(v_0) = 1$ and put $W = \ker \alpha$. Let $t_0 < \dots < t_m$ be the distinct vertex heights, all in R . Between consecutive heights, intersect each input cell P with the height hyperplane and translate the section by $-tv_0$ into W . Give the section the orientation for which height first recovers the orientation of P .

Let L_P be the rational tangent space of P , let β_P be its coefficient, and choose $w_P \in L_P$ with $\alpha(w_P) = 1$. Then

$$(5.4) \quad \gamma_P = \iota_\alpha \beta_P \in \bigwedge^{q-1} (L_P \cap W) \wedge \bigwedge^{p-q} W, \quad \beta_P = w_P \wedge \gamma_P.$$

To verify the first assertion, extend a basis of $L_P \cap W$ by w_P to a basis of L_P and then extend the horizontal basis to W . Factoring the tangent wedge of P from β_P and contracting removes w_P and leaves all the horizontal tangent factors. The second assertion follows by contracting $w_P \wedge \beta_P = 0$. Thus the sum of these translated sections with coefficients γ_P is a chain $\zeta_j(t)$ of type $(q-1, p-1)$ in W .

At a noncritical height, truncate z below t . Rational height cuts give admissible cells over $R + \mathbb{Q}t$. To justify the boundary cancellation, first choose a finite common admissible refinement of the original facets and then apply the same height cut to each occurrence. The original facet

terms cancel by $\partial z = 0$. The remaining boundary is the height section retaining the full β_P coefficients. Apply ∂ once more and contract by α . Boundary retains the coefficient and therefore commutes with this contraction. After translation into W we obtain

$$\partial \zeta_j(t) = 0.$$

Every face direction of the section is rational: it is the intersection of a rational face direction with W . Each section vertex lies on an input edge with an endpoint x_* and rational direction u , and has the formula

$$x(t) = x_* + \frac{t - \alpha(x_*)}{\alpha(u)} u.$$

Since $\alpha(u)$ is a nonzero rational number, translation by $-tv_0$ writes this vertex as $a + tb$, with $a \in R \otimes W$ and $b \in W$. On an open interval between vertex heights, the intersected faces are fixed. Their rational carrier equations hold identically in these affine vertex formulas.

Choose $\tau_j \in (t_j, t_{j+1})$ independent of R and set $R_j = R \oplus \mathbb{Q}\tau_j$. The slice $\zeta_j(\tau_j)$ is a finite closed chain in $C_{q-1,p-1}(R_j; W)$. The inequalities

$$0 < q - 1 < p - 1$$

permit the outer induction, which gives a finite chain

$$\widetilde{K}_j \in C_{q,p-1}(R_j; W), \quad \partial \widetilde{K}_j = \zeta_j(\tau_j).$$

With $e_t^j : R_j \rightarrow R$ fixing R and sending τ_j to t , Theorem 4.3 gives

$$(5.5) \quad \partial \operatorname{Sp}_{e_t^j} \widetilde{K}_j = \operatorname{Sp}_{e_t^j} \zeta_j(\tau_j) \quad (t \in R).$$

This is an identity in the finite admissible-chain quotient. It remains valid through any folds or rank collapse of the filling cells.

Step 2. We prove that endpoint specializations of neighboring slices agree at every critical height. Within the open interval, the specialized slice on the right of (5.5) is the actual geometric slice: its affine vertex formulas preserve the face lattice, and its barycentric domain maps onto that convex section with the chosen orientation. We also need the endpoint specializations to agree across adjacent intervals.

For one original convex cell, evaluate the finitely many affine section-vertex formulas at an endpoint height. Its face-barycentric domain then has a finite piecewise affine map, allowing repeated vertices and collapsed simplices. Integration of a smooth form over each image simplex is continuous in its vertex coordinates. A simplex that loses rank has zero integral in the original dimension. Meanwhile the actual convex sections tend to the critical convex section. If the latter has dimension $q - 1$, its fundamental-chain integrals tend to those of that section: in coordinates in its moving carrier, changes occur in shrinking neighborhoods of the finitely many limiting faces. These neighborhoods have volume tending to zero. If the critical section has smaller dimension, the section volumes themselves tend to zero. It follows that the finite specialized chain is precisely the critical section, or zero in dimension $q - 1$ when the section collapses.

This identification is an equality of admissible finite chains. Indeed, a finite common rational refinement separates their coefficients, and a test form supported near the relative interior of one refined cell detects its coefficient. Equality of the integration functionals therefore detects equality in the subdivision quotient. The finite simplex argument identifies the endpoint chains; it is not used to obtain a filling by a limit.

No positive-dimensional face of an input cell is horizontal, because such a face contains an edge. Its minimum and maximum height faces are consequently single vertices. Since $q \geq 2$, these extreme sections vanish in degree $q - 1$. At every other critical height the two one-sided chains are the same oriented section. Summing over the input cells proves equality of adjacent slice specializations, with zero slice chains at the two extreme heights.

Step 3. We match the higher-degree slice fillings and terminate the family at zero. Starting with prescribed lower chain zero, apply the correction (5.2) in degree q . More explicitly, if K_* is the chain prescribed at height t_j , regard $K_* - \text{Sp}_{e_{t_j}^j} \widetilde{K}_j$ as a constant chain over R_j and put

$$K_j = \widetilde{K}_j + K_* - \text{Sp}_{e_{t_j}^j} \widetilde{K}_j.$$

The endpoint slice agreement shows that the correction is closed. Thus $\partial K_j = \zeta_j(\tau_j)$, its lower specialization is K_* , and its upper specialization prescribes the next K_* . The chains themselves are matched at every junction, including all their coefficients and subdivision relations.

The final K_* is closed in $C_{q,p-1}(R; W)$, since the final slice is zero. We have $1 \leq q \leq p-1 \leq \dim W$. Apply Lemma 5.1, with $k = q$ and $s = p-1$, to append closed families from K_* to zero. The lemma uses specialization to realize each parameter-elimination family at an independent reference height. Functoriality identifies its endpoint chains even if the original cells fold during elimination. Its positive interval lengths and all endpoint heights belong to R .

Step 4. We sweep the matched fillings and recover every input band with its full coefficient. Apply Theorem 4.5 to every K_j and every terminal family. Their horizontal coefficient degree is $p-1$, so the resulting sweeps belong to $C_{q+1,p}(R; U)$. For a cell of horizontal direction L and coefficient δ , the lifted coefficient is $(v_0 + b) \wedge \delta$, with b an anchor velocity. It contains the entire swept direction $L + \mathbb{Q}(v_0 + b)$. Independence of the anchor and linearity on the subdivision quotient are supplied by Theorem 4.5. In particular they preserve the coefficient cancellations in the boundary identity for K_j .

We check that sweeping $\partial K_j = \zeta_j(\tau_j)$ gives the corresponding band of z , with its full coefficient. For the slice of an original cell P , its anchor track can be taken along an original edge. Its rational velocity in U is therefore a vector $\widehat{w}_P \in L_P$ of height one. Contracting $\widehat{w}_P \wedge \beta_P = 0$ gives

$$\widehat{w}_P \wedge \gamma_P = \beta_P.$$

Thus the sweep coefficient is the original β_P .

The geometric band is $P \cap \{t_j \leq \alpha \leq t_{j+1}\}$ with the original orientation. Its upper and lower caps agree with the finite specialized slices just identified. Its side faces agree with the sweeps of their sections, by induction on face dimension; in dimension one this is the literal oriented edge track. The height first boundary convention is upper cap minus lower cap minus the swept boundary of the horizontal section, exactly as in Theorem 4.5. Thus the band and the constructed sweep have the same boundary in their common rational q -dimensional affine carrier.

Two compact finite top-dimensional chains in one affine space of positive dimension with the same boundary agree. To see this, take the difference and cut the space by the finitely many supporting hyperplanes of its facets. Its multiplicity is constant on every open chamber. Zero boundary says that the multiplicities agree across every wall, componentwise in the coefficient space. A generic polygonal path connects a chamber to the complement of the compact support, so this multiplicity is zero. This proves the asserted equality of the sweep and the band as finite chains. Applied cellwise, it identifies the side sweeps of the original interval fillings with all of z . The closed terminal families have zero side sweeps.

Step 5. We reduce the remaining cap residual to horizontal cycles and fill them by the dimension induction. Let $\mathcal{D}$ be the sum of all these finite sweeps. Summing their boundary formulas gives

$$\partial \mathcal{D} = \rho - z,$$

where ρ retains the full cap differences on all interval boundaries, including the extreme caps. Contraction of each cap recovers its specialized horizontal filling. Exact matching and the zero extreme chains imply that $\iota_\alpha \rho = 0$ on each height level. No equality of the full adjacent caps is required.

After a common admissible refinement, a residual q -cell has rational direction $L \subset W$ and coefficient $\eta \in \bigwedge^q L \wedge \bigwedge^{p-q} U$. Decompose it as

$$\eta = \eta_0 + v_0 \wedge \eta_1, \quad \eta_0 \in \bigwedge^p W, \quad \eta_1 \in \bigwedge^{p-1} W.$$

Contraction gives $\eta_1 = 0$. Extending a basis of L to one of W shows that incidence is retained in the horizontal component:

$$\eta = \eta_0 \in \bigwedge^q L \wedge \bigwedge^{p-q} W.$$

Hence every level of ρ , translated by its height into W , belongs to $C_{q,p}(R; W)$. It is closed, since $\partial\rho = 0$ follows from $\partial^2\mathcal{D} = 0$ and $\partial z = 0$, and different height levels are disjoint.

The inner induction, at the same q, p and in dimension $n - 1$, fills each translated level in W . If $n - 1 < p$, the residual coefficient space vanishes. Restoring the heights and summing the fillings gives a finite $\mathcal{F} \in C_{q+1,p}(R; U)$ with $\partial\mathcal{F} = \rho$. Consequently

$$\partial(\mathcal{F} - \mathcal{D}) = z.$$

The outer induction applies only to the slice degree $q - 1$, with coefficient degree $p - 1$; the inner induction keeps q, p fixed and decreases the ambient dimension. Each branch reaches $q = 1$ or a zero coefficient space. There are finitely many slices and terminal intervals at each stage, and all specializations and sweeps have finite output. Their endpoints lie in R , and any rectification or common refinement solves rational linear equations with constants in R . Every final vertex therefore lies in $R \otimes U$. Rational coefficients and rational face directions are preserved. This completes both inductions and proves the theorem. $\square$

6. PERIODIC GEOMETRY AND STRAIGHT CYCLES

We retain the notation of Set-up 2.1, and write $\pi : V_{\mathbb{R}} \rightarrow X$ for the quotient map. We first construct a periodic polyhedral decomposition compatible with any finite list of cells. We then identify closed incidence chains with balanced straight cycles and compare their full marked classes. The construction uses bounded coordinate neighborhoods, so it applies to arbitrary real period lattices.

Theorem 6.1. *In Set-up 2.1, let $P_1, \dots, P_m$ be any finite list of bounded convex rational-faced polytopes with vertices in A . There is a Γ -periodic locally finite face-to-face decomposition of $V_{\mathbb{R}}$ into bounded convex rational-faced polytopes with vertices in A and finitely many cell orbits, such that every $P_i + \gamma$, for $\gamma \in \Gamma$, is a finite union of its cells. Every cell of this decomposition projects injectively to X . The list may be empty.*

Thus a finite collection of cells can be refined on the original torus, even when its projections overlap or wind around the torus. The resulting decomposition is polytopal; no rational triangulation is required.

Proof. We embed the torus by finitely many periodic piecewise affine functions, refine their image by one finite hyperplane arrangement, and lift the resulting cells through explicit affine inverses.

Step 1. We verify that rational carrier and facet equations preserve the chosen parameter space. Use the fixed basis to identify $V_{\mathbb{R}}$ with $\mathbb{R}^d$ and A with S^d . A rational-faced polytope with vertices in S^d admits affine carrier equations and facet inequalities with rational linear parts and constants in S . For the carrier, take rational covectors annihilating its tangent space and evaluate them at one vertex. For a facet, extend a rational defining covector on the carrier to $\mathbb{Q}^d$ and evaluate it at a facet vertex. Conversely, a bounded polyhedron defined by such equations and inequalities has rational face directions and vertices in S^d . Indeed, at a vertex the active equations, including the carrier equations, contain d independent rows. Their inverse matrix is rational. The same observation holds in every ambient dimension and for bounded intersections. It uses only rational linear combinations of elements of S .

Step 2. We construct periodic piecewise affine coordinates that embed the torus and have explicit local affine inverses. Discreteness gives a number $\lambda > 0$ such that $\|\gamma\|_\infty \geq \lambda$ for every nonzero $\gamma \in \Gamma$. Choose $h \in S$ with $0 < h < \lambda/10$. This is possible because S contains a nonzero element and its rational multiples are dense in $\mathbb{R}$. In particular, A is dense in $V_{\mathbb{R}}$. Compactness of X supplies $c_1, \dots, c_\ell \in A$ such that the images of $c_a + (-h/2, h/2)^d$ cover X . Put

$$D_a = c_a + [-h/2, h/2]^d, \quad B_a = c_a + [-4h, 4h]^d.$$

For fixed a , distinct translates of B_a by Γ are disjoint: an intersection would give a nonzero period of sup norm at most $8h < \lambda$. Thus each B_a projects injectively.

For $t \in \mathbb{R}^d$, define

$$b(t) = \max\{0, \min\{h, 4h - \|t\|_\infty\}\}, \quad g_i(t) = \min\{b(t), \max\{-b(t), t_i\}\} \quad (1 \leq i \leq d).$$

These continuous functions vanish on and outside the boundary of $[-4h, 4h]^d$. On $[-h/2, h/2]^d$ they satisfy $b = h$ and $g_i = t_i$. They are finite compositions of minima and maxima of affine functions with integral slopes and constants in S . On $B_a + \gamma$ use $b(x - c_a - \gamma)$ and $g_i(x - c_a - \gamma)$, and put these functions equal to zero outside the union of those boxes. This defines continuous periodic functions b_a, g_{ai} on $V_{\mathbb{R}}$. Their supports are locally finite. Their affine pieces have rational slopes and constants in S , since $c_a, \gamma \in A$.

Set $M = \ell(d+1)$ and define

$$\Phi : X \longrightarrow \mathbb{R}^M, \quad \Phi = (b_a, g_{a1}, \dots, g_{ad})_{1 \leq a \leq \ell}.$$

We also write Φ for its periodic lift. To prove injectivity, choose a lift $\tilde{x} \in D_a$ of $x \in X$. Its a th block satisfies $b_a(x) = h$ and $g_{ai}(x) = (\tilde{x} - c_a)_i \in [-h/2, h/2]$. If $\Phi(y) = \Phi(x)$, then $b_a(y) = h > 0$ gives a unique lift $\tilde{y} \in B_a$. The clipped coordinate $\min\{h, \max\{-h, (\tilde{y} - c_a)_i\}\}$ has value strictly between $-h$ and h . Consequently it equals $(\tilde{y} - c_a)_i$ itself. Equality of all these coordinates gives $\tilde{y} = \tilde{x}$. The continuous injection Φ is an embedding because its domain is compact.

Write the target coordinates as $z_{a0}, z_{a1}, \dots, z_{ad}$, and put

$$(6.1) \quad \rho_a(z) = c_a + (z_{a1}, \dots, z_{ad}).$$

Then $\rho_a(\Phi(x)) = x$ on D_a . On $D_a + \gamma$, the inverse lift is $\rho_a + \gamma$. Wherever Φ is affine on a k -dimensional polytope in one of these boxes, its restriction has rank k , and the displayed coordinate projection is its affine inverse on the image. Both maps have rational linear parts and constants in the appropriate power of S .

Step 3. We construct a finite list of image polytopes recording all prescribed cells in every chart. Only finitely many boxes $B_b + \gamma$ meet a fixed D_a : for each b the possible periods lie in the bounded set $D_a - B_b$. Cut D_a by the supporting hyperplanes of these boxes and by the comparison hyperplanes between the affine functions in their minimum and maximum formulas. There are finitely many such hyperplanes, all with rational normals and constants in S . On each resulting cell the selected branches are affine; continuity gives the same assertion on its closure. Include all faces, and list the resulting polytopes from all the boxes as $R_1, \dots, R_t$. Their projections cover X . Each R_ν lies in some D_a , and $\Phi|_{R_\nu}$ is affine and injective, with inverse ρ_a on its image. Therefore $\Phi(R_\nu)$ is a bounded convex rational-faced polytope with vertices in S^M .

For every ν, i , there are only finitely many $\gamma \in \Gamma$ for which $R_\nu \cap (P_i + \gamma)$ is nonempty. Include the images of all these intersections and their faces in the image list. The intersections are rational-faced with vertices in A by the rational-equation observation. Their images have rational face directions and vertices in S^M , because Φ has the affine inverse just described on R_ν . Thus we have a finite list of polytopes in $\mathbb{R}^M$ covering $\Phi(X)$ and recording the prescribed cells in every chart.

Step 4. We refine the image polytopes by one finite hyperplane arrangement in the target. Choose a coordinate box with endpoints in S containing the entire image list in its interior. Such endpoints can be chosen as rational multiples of any positive element of S . For every polytope in the list, take rational equations for its affine carrier and rational facet inequalities within

that carrier. Extend the latter covectors rationally to $\mathbb{R}^M$. All their constants lie in S . Cut the bounding box by these finitely many hyperplanes and include all faces. This gives a finite face-to-face polytopal complex with rational face directions and vertices in S^M .

Each listed polytope is a union of arrangement cells: the signs of all its defining equations and inequalities are fixed on the relative interior of each cell. Retain precisely the cells contained in $\Phi(X)$ and call the resulting subcomplex $\mathcal{K}$. It is a subcomplex because $\Phi(X)$ is the union of the closed polytopes $\Phi(R_\nu)$, each already a union of arrangement cells. Every cell Q of $\mathcal{K}$ is contained in some $\Phi(R_\nu)$. Indeed, choose a point in the relative interior of Q and an image $\Phi(R_\nu)$ containing it. The arrangement signs put the whole relative interior in this image, and closedness then puts all of Q in it. Hence Φ^{-1} on Q has the affine lift $\rho_a|_Q$ from (6.1) for a box containing R_ν .

Step 5. We lift the target cells to the required periodic face-to-face decomposition on the original lattice. For each $Q \in \mathcal{K}$, choose one such lift $\sigma_Q = \rho_a(Q) \subset D_a$. It is a bounded convex rational-faced polytope with vertices in A , and it projects injectively. Its projection is exactly $\Phi^{-1}(Q)$. Take all translates $\sigma_Q + \gamma$, with $\gamma \in \Gamma$.

We check their compatibility, including along lower-dimensional faces. If U is a face of Q , the restriction $\rho_a|_U$ and the chosen lift of $\Phi^{-1}(U)$ differ pointwise by an element of Γ . Their difference is continuous on the connected polytope U , so it is one constant period. Thus every face occurs in the translated collection. If two target cells meet, their intersection is a common face U . The restrictions to U of any two chosen translated lifts again differ by one constant period. If the lifted cells meet at one point, this period is zero, and their intersection is the entire lift of U , a face of both. Otherwise they are disjoint. This proves the face-to-face property upstairs.

The lifted cells cover $V_{\mathbb{R}}$ because the target cells cover $\Phi(X)$. There are finitely many orbits. Local finiteness follows from boundedness and discreteness: a compact set can meet a translate of σ_Q only for periods in its bounded difference with σ_Q .

Finally, fix P_i . Only finitely many translates $R_\nu + \delta$ meet it, and those translates cover it. The intersection $P_i \cap (R_\nu + \delta)$ translates back to $R_\nu \cap (P_i - \delta)$, whose image was included in the finite list. That image is a finite union of cells of $\mathcal{K}$. Applying its own affine inverse, and translating back by δ , expresses the actual intersection upstairs as a union of lifted cells. Each such lift differs from the chosen one by a constant element of Γ , so it is present in our collection. This proves that P_i itself, and hence every $P_i + \gamma$, is a finite union of cells. The construction also works when the input list is empty. All vertices remain in A , and all translations belong to the original Γ . $\square$

We use the straight-cycle class construction of Mikhalkin and Zharkov [MZ14, Section 4.1, Definition 14 and Proposition 11, pp. 328–329].

Theorem 6.2 (Mikhalkin–Zharkov). *An integer weighted straight subcycle of dimension r in a tropical space has a tropical homology class of bidegree (r, r) , represented by its oriented top cells with coefficient equal to the weight times the primitive integral tangent exterior generator. The straight-subcycle weights may have either sign.*

The framing in this statement is attached to the saturated tangent lattice of each top cell. It is this framing that converts an incidence coefficient into the scalar weight of a straight cycle.

Definition 6.3. In Set-up 2.1, let $1 \leq r \leq d$ and $0 \leq q \leq r$. For a finite chain $c \in C_{q,r}$, its *periodization* is

$$\mathcal{P}_\Gamma(c) = \sum_{\gamma \in \Gamma} T_\gamma c,$$

viewed as a locally finite oriented polyhedral chain with coefficients in $\bigwedge^r V$. Two such geometric chains agree when, after common local subdivisions and a choice of cell orientations, their coefficients agree on every q -dimensional relative interior.

Boundedness of the cells makes the sum locally finite. Translation relations have zero periodization, and we will verify that finite subdivision relations give precisely the expected geometric equalities.

Proposition 6.4. *In Set-up 2.1, let $1 \leq r \leq d$. Periodization is injective on $(C_{q,r})_\Gamma$ for $0 \leq q \leq r$. In top degree it identifies closed coinvariant chains with the finite rational signed straight tropical r -cycles having lifted vertices in A , modulo finite compatible refinements. Under this identification, all signed occurrence multiplicities are preserved, and the ordinary marked class is exactly the primitive-framed tropical homology class in $\bigwedge^r G \otimes \bigwedge^r V$. More generally, a finite rational signed cycle with arbitrary real offsets becomes an integer straight cycle after multiplication by one positive integer, and its rational class is the corresponding integer straight class divided by that integer.*

Thus the geometric realization retains the whole marked tensor class, including the multiplicities of repeated occurrences of the same cell.

Proof. Put $B = \bigwedge^r V$. We first recover coinvariant chains from their periodic coefficients. We then identify the boundary equation with primitive balancing and compare the resulting framed homology chains.

Step 1. We prove that periodization is a well-defined injective chain map on ordinary coinvariants. The interpretation of a finite incidence chain as a geometric chain is faithful. For $q > 0$, group a finite presentation by its affine q -dimensional carriers and orient each carrier. Within one carrier, cut the polytopes by all the finitely many facet hyperplanes occurring there. These are admissible subdivisions: their equations have rational linear parts and constants in S , so their vertices lie in A by the rational-equation argument in the preceding proof. On each resulting q -cell there is one summed coefficient in $\bigwedge^q L \wedge \bigwedge^{r-q} V$, where L is its rational tangent space. If the geometric chain is zero, every such sum is zero, and coefficient linearity makes the incidence chain zero. Distinct q -dimensional carriers meet in dimension less than q , so they cannot cancel a coefficient on a relative open q -cell. For $q = 0$ the assertion is the independence of the coefficients at distinct points.

Internal facets of any finite subdivision cancel with opposite orientations and identical retained coefficients. It follows that the geometric boundary agrees with the incidence boundary. Periodization respects these subdivisions and commutes with boundary, since the same calculation applies on each bounded neighborhood. Reindexing its locally finite sum gives $\mathcal{P}_\Gamma(T_\delta c - c) = 0$ for $\delta \in \Gamma$. Thus it defines a chain map on ordinary coinvariants.

To prove injectivity, choose a finite cell presentation of c and apply Theorem 6.1 to its cells. Subdivide each occurrence into the q -cells of the resulting periodic complex. For each cell orbit $\mathcal{O}$ occurring in this finite presentation, choose a representative $\sigma_{\mathcal{O}}$, orient it, and translate that orientation throughout its orbit. Let $\beta_{\mathcal{O}} \in B$ be the sum of all the coefficients in this orbit, including the signs from changes of orientation. In coinvariants we obtain

$$\bar{c} = \sum_{\mathcal{O}} \sigma_{\mathcal{O}} \otimes \beta_{\mathcal{O}}.$$

A nonempty bounded cell has no nonzero translation stabilizer. If $\sigma + \gamma = \sigma$, then iterating the translation keeps $x + n\gamma$ in the bounded set σ for all integers n , forcing $\gamma = 0$. Consequently the coefficient of $\mathcal{P}_\Gamma(c)$ on $\sigma_{\mathcal{O}}$ is exactly $\beta_{\mathcal{O}}$. If periodization vanishes, every orbit sum is zero, so $\bar{c} = 0$. This also proves injectivity for points.

In particular, a cell which winds around the torus contributes once for every occurrence in each refined orbit. A rational affine carrier may have a nontrivial period stabilizer, but the bounded cells used in this calculation do not. No division by a carrier stabilizer enters the orbit sums.

Step 2. We identify closed top-degree chains with rational signed cycles satisfying primitive balancing. Let $\bar{c}$ be closed in $(C_{r,r})_\Gamma$. Refine a finite representative as above. Its periodic

geometric boundary vanishes. For an oriented top cell P , write L_P for its rational tangent space and put

$$N_P = N \cap (L_P \otimes \mathbb{R}).$$

This is a saturated rank- r lattice: if a nonzero integer multiple of $v \in N$ belongs to $L_P \otimes \mathbb{R}$, then v belongs to that space too. Let ξ_P be its oriented primitive exterior generator. The top coefficient lies in the rational line $\bigwedge^r L_P$, so it has a unique expression $w_P \xi_P$ with $w_P \in \mathbb{Q}$. Reversing the orientation changes both the oriented cell and ξ_P by a sign; it leaves w_P unchanged. Sum coincident contributions, remove the zero-weight top cells, and retain the faces of the remaining cells.

Fix an oriented facet F and write η_F for the primitive exterior generator of $N_F = N \cap (L_F \otimes \mathbb{R})$ in that orientation. For each incident P , let $s(P, F) \in \{1, -1\}$ be the boundary incidence sign. The lattice N_F is saturated in N_P , and the rank-one quotient N_P/N_F is saturated in N/N_F . For the latter assertion, if kv modulo N_F lies in N_P/N_F , then $kv \in N_P$; saturation of N_P gives $v \in N_P$. Hence the primitive generator of N_P/N_F is also primitive in the ambient quotient lattice.

Let $u_{P/F} \in N_P$ lift this generator pointing from F into P . An oriented basis of N_F , followed by this lift, is a basis of N_P . The outward-normal-first boundary convention therefore gives

$$(6.2) \quad s(P, F) \xi_P = (-1)^r \eta_F \wedge u_{P/F}.$$

Indeed, the outward vector is $-u_{P/F}$, so the top exterior generator is $(-u_{P/F}) \wedge s(P, F) \eta_F$; moving the first vector to the last position gives the displayed identity. There is no lattice index, because the vectors just used form a lattice basis.

The retained coefficient at F in the boundary is consequently

$$\sum_{P \supset F} s(P, F) w_P \xi_P = (-1)^r \eta_F \wedge \sum_{P \supset F} w_P u_{P/F}.$$

The map

$$V/L_F \longrightarrow \bigwedge^r V, \quad [v] \longmapsto \eta_F \wedge v$$

is injective: extend a basis of L_F to a basis of V and inspect the distinct exterior basis elements. This includes $r = 1$, when $\eta_F = 1$ and $L_F = 0$. Thus boundary zero is exactly

$$\sum_{P \supset F} w_P u_{P/F} = 0 \quad \text{in } V/L_F,$$

with the inward normals of Definition 2.4. The result is a finite rational signed straight cycle with vertices in A .

Conversely, for such a cycle choose one bounded lift and an orientation for each top-cell orbit and attach $w_P \xi_P$. This gives a finite coinvariant chain, independent of the chosen lifts. Balancing and (6.2) make its periodized boundary zero. Injectivity in degree $r - 1$ then makes its coinvariant boundary zero. The two constructions are inverse by the orbit-sum calculation. Applying Theorem 6.1 to the union of two finite lists of representatives provides a common refinement, so these constructions respect precisely the stated equivalence of presentations.

Step 3. We compare rational signed cycles with integer straight cycles and identify their ordinary framed homology classes. Every sufficiently small chart of X lifts to an open subset of $V_{\mathbb{R}}$, and the transition maps are translations with derivative id on N . A rational-faced polytope with real offsets is locally described by rational affine carrier equations and rational facet inequalities. Clearing their denominators gives integral normals and real constants, the polyhedral domains used for straight subcycles. In these charts the nonzero weighted support of the cycle has a finite rational polyhedral local model, and its primitive-normal balancing is the weighted balancing of that model.

Let Z be a finite rational signed cycle, allowing arbitrary real offsets. Choose one positive integer D clearing its finitely many orbit weights. Then DZ has integer weights and the same balancing equations, so it is an integer straight cycle to which Theorem 6.2 applies. Neither

the positions of its cells nor the period lattice change in this operation. If an auxiliary injective refinement is needed for these arbitrary offsets, adjoin the finitely many vertex coordinates of orbit representatives to S and apply Theorem 6.1 with that finite parameter space. For a cycle already having vertices in A , the original S suffices throughout.

The ambient tropical coefficient system in degree r is constant with fiber $\bigwedge^r N$: every ambient tangent lattice is N , and all transition derivatives are the identity. After rationalization it is the constant system B . Choose an ordinary topological triangulation subordinate to the refined cells and the lifted charts. One can obtain such a triangulation by barycentrically subdividing the finite target complex in the preceding proof and pulling its simplices back by the local affine inverses. Orient each top simplex by its containing cell. This is only an ordinary singular-chain construction; its diagonals need not have rational directions.

On a simplex contained in P , the class construction in Theorem 6.2 for DZ assigns the coefficient $Dw_P\xi_P$. Dividing by D gives exactly the incidence coefficient $w_P\xi_P$. Internal simplex facets cancel in pairs; on the remaining facets, cancellation is the balancing identity already proved. Thus both constructions give the same closed ordinary B -valued chain. The auxiliary embedding is used only to choose compatible cells and simplices. The tangent framing is always taken in the original lattice N , and the coefficients are retained in B .

Finite subdivisions preserve this ordinary class. For a subdivided convex cell, a compatible ordinary triangulation expresses its oriented relative fundamental chain as the sum of those of the pieces. Changing triangulations gives homologous relative chains: subdivide both, and use the ordinary prism subdivision identity, compatibly on their faces. Applying this with the constant coefficient on each cell proves invariance for the closed chain. Translation by an original period projects to the same parametrized piece on X . These observations also show that the class is independent of the refinement and equals the ordinary class of the original coinvariant chain.

Step 4. We compare every coordinate of the full marked tensor class using the positive-period basis. Choose an integral basis $\gamma_1, \dots, \gamma_d$ of Γ , and let Π be its real column matrix in the fixed tangent basis. The coordinates $y = \Pi^{-1}x$ identify X with $\mathbb{R}^d/\mathbb{Z}^d$. Its positively parameterized coordinate subtori give the usual identification

$$H_r(X; B) = \bigwedge^r G \otimes B.$$

For each increasing r -tuple $I = (i_1, \dots, i_r)$, the form $dy_I = dy_{i_1} \wedge \dots \wedge dy_{i_r}$ integrates to 1 on its oriented coordinate subtorus and to 0 on the other coordinate r -subtori. Pairing these forms with a dual basis of B detects every coordinate of the displayed tensor product. The evaluations on a refined representative are

$$\sum_{P \bmod \Gamma} w_P \beta^*(\xi_P) \int_P \pi^*(dy_I), \quad \beta^* \in B^*.$$

Each integral is additive under subdivision, and its coefficient is the actual signed orbit sum. The ordinary class and the straight class therefore agree in every coordinate of the full marked tensor product. For arbitrary real offsets the same comparison gives $\text{cl}(Z) = D^{-1}[DZ]$. This proves all the asserted class identities and multiplicity statements. $\square$

7. THE CYCLE-CLASS IMAGE

We retain Set-up 2.1. We first turn the endpoint lift into a closed chain on the torus and compute its complete marked class. We then prove the converse by applying the eigenwave to each tangent volume vector. Throughout the class calculation, vector volumes retain the exterior coefficient as a separate tensor factor.

Proposition 7.1. *In Set-up 2.1, fix $1 \leq r \leq d$, set $B = \bigwedge^r V$, and let h be the marked Koszul differential using $T_{\gamma_i} - 1$ for an integral basis of Γ . Let $w \in \ker W_r$. There are finite chains*

$$a_j \in \bigwedge^{r-j} G \otimes C_{j,r} \quad (0 \leq j \leq r), \quad \partial a_j = h a_{j-1} \quad (1 \leq j \leq r),$$

with coefficient sum $I_0(a_0) = w$. For every such collection, if I_q denotes the ordinary oriented vector integral retaining the coefficient in B , then

$$I_r(a_r) = (-1)^{r(r-1)/2} \left(\bigwedge^r j_{\mathbb{R}} \otimes \text{id} \right) w.$$

The periodization of $(-1)^{r(r-1)/2} a_r$, refined as in Proposition 6.4, has exact marked class w .

Thus the finite affine fillings produce the prescribed rational class on the original torus. The sign is determined by ordinary integration with the positive period marking.

Proof. We construct the finite staircase, compute its vector integral by Stokes, and identify the resulting class by constant forms on the torus.

Step 1. We construct the finite staircase on the original integral period lattice. Choose the indicated integral basis $\gamma_1, \dots, \gamma_d$ of Γ . For an increasing tuple $I = (i_1, \dots, i_p)$, write $\gamma_I = \gamma_{i_1} \wedge \dots \wedge \gamma_{i_p}$ and $I \setminus i_a = (i_1, \dots, \widehat{i_a}, \dots, i_p)$. On chains with coefficients in $C_{q,r}$ the differential is

$$h(\gamma_I \otimes c) = \sum_{a=1}^p (-1)^{a-1} \gamma_{I \setminus i_a} \otimes (T_{\gamma_{i_a}} - 1)c.$$

Translations commute with the facet boundary, so $h\partial = \partial h$. They also commute with one another. In h^2 , the terms deleting any two distinct indices therefore cancel with opposite signs, giving $h^2 = 0$.

Let $M = C_{0,r}/\partial C_{1,r}$, and let $\pi : C_{0,r} \rightarrow M$ be the quotient map. Coefficient sum on endpoint chains descends to a map $\epsilon : M \rightarrow B$, since the two endpoints of an oriented segment have the same retained coefficient and opposite boundary signs. Theorem 3.3, with $(U, R, p, \Delta, a) = (V, S, r, \Gamma, r)$, gives a Koszul cycle

$$z \in \bigwedge^r G \otimes M, \quad hz = 0, \quad (\text{id} \otimes \epsilon)z = w.$$

Choose a finite endpoint representative for each of its finitely many coefficients. Their sum is a_0 , with $(\text{id} \otimes \pi)a_0 = z$ and $I_0(a_0) = w$. Every coefficient of ha_0 has zero image in M . Theorem 3.2 therefore fills these coefficients by finite incident segment chains. This gives a_1 with $\partial a_1 = ha_0$.

Suppose $2 \leq q \leq r$ and a_{q-1} has been constructed. Then

$$\partial(ha_{q-1}) = h\partial a_{q-1} = h^2 a_{q-2} = 0.$$

Expand ha_{q-1} in the basis γ_I of $\bigwedge^{r-q} G$. Each coefficient is a finite closed chain of degree $q-1$, and $0 < q-1 < r$. Theorem 5.3 fills each coefficient in $C_{q,r}$. These finitely many fillings give a_q with $\partial a_q = ha_{q-1}$. When $r = 1$, the segment construction already completes the staircase. Every invocation of the filling theorems retains rational coefficients, rational face directions, and vertices in A . All translations used in h belong to the original Γ .

Step 2. We derive the Stokes recurrence that transfers one period at each stage of the staircase. Use the fixed basis $v_1, \dots, v_d$ of N and its real linear coordinates $x_1, \dots, x_d$. For $0 \leq q \leq d$ define the $\bigwedge_{\mathbb{R}}^q V_{\mathbb{R}}$ -valued constant differential form

$$\Omega_q = \sum_{i_1 < \dots < i_q} (v_{i_1} \wedge \dots \wedge v_{i_q}) dx_{i_1} \wedge \dots \wedge dx_{i_q}, \quad \Omega_0 = 1.$$

For a cell P carrying $\beta \in B$, set

$$I_q(P \otimes \beta) = \left(\int_P \Omega_q \right) \otimes \beta \in \bigwedge_{\mathbb{R}}^q V_{\mathbb{R}} \otimes_{\mathbb{Q}} B.$$

Extend this linearly and as the identity on any additional $\wedge^p G$ factor. In degree zero this is coefficient sum. In degree one, $I_1([x, y] \otimes \beta) = (y - x) \otimes \beta$. Integration respects orientation reversal and finite subdivision, so it is defined on the incidence chains used above.

Write $x = \sum_i x_i v_i$, and wedge x into the vector factor of Ω_{q-1} . Direct differentiation gives

$$(7.1) \quad d(x \wedge \Omega_{q-1}) = q\Omega_q \quad (q \geq 1).$$

For each increasing q -tuple, there are q choices of the vector contributed by differentiating x . Reordering that vector and its differential into the increasing tuple gives the same sign in both factors, so each choice contributes $+\Omega_q$ in that coordinate. Stokes and (7.1), with the exterior coefficient retained, now yield

$$(7.2) \quad I_q(c) = \frac{1}{q} \int_{\partial c} x \wedge \Omega_{q-1}.$$

Here and below the integral of a vector-valued form on a chain retains its coefficient in B . For $\gamma \in \Gamma$ and a finite $(q-1)$ -chain b , translation invariance of Ω_{q-1} gives

$$(7.3) \quad \int_{(T_{\gamma-1})b} x \wedge \Omega_{q-1} = j(\gamma) \wedge I_{q-1}(b).$$

Indeed, pullback by translation replaces x by $x + j(\gamma)$, and subtraction cancels the term containing x . These applications of Stokes may be made on ordinary real triangulations of the finitely many polytopes. Such triangulations serve only to integrate and do not alter the admissible subdivisions used in constructing the chains.

Define a rational-linear operator on $\wedge^p G \otimes \wedge_{\mathbb{R}}^q V_{\mathbb{R}} \otimes B$ by

$$\kappa(\gamma_I \otimes \eta \otimes \beta) = \sum_{a=1}^p (-1)^{a-1} \gamma_{I \setminus i_a} \otimes (j(\gamma_{i_a}) \wedge \eta) \otimes \beta.$$

It lowers the period degree by one and raises the vector-volume degree by one. Apply (7.2) to each coefficient of a_q , use $\partial a_q = h a_{q-1}$, and evaluate each translation difference by (7.3). The result is

$$(7.4) \quad I_q(a_q) = \frac{1}{q} \kappa I_{q-1}(a_{q-1}) \quad (1 \leq q \leq r).$$

Step 3. We compute the staircase sign, cancel the factorial, and recover the complete prescribed marked class. Consider $\kappa^r(\gamma_I \otimes 1 \otimes \beta)$ for $|I| = r$. Its terms correspond to the $r!$ orders in which the indices can be removed. For an order specified by a permutation σ , the product of the removal signs is $\text{sgn}(\sigma)$. Each transferred vector is wedged on the left. The final vector wedge therefore has the reverse of the removal order, whose sign relative to increasing order is $(-1)^{r(r-1)/2} \text{sgn}(\sigma)$. The two signs multiply to $(-1)^{r(r-1)/2}$ for every order. Consequently

$$\kappa^r(\gamma_I \otimes 1 \otimes \beta) = r! (-1)^{r(r-1)/2} (j(\gamma_{i_1}) \wedge \cdots \wedge j(\gamma_{i_r})) \otimes \beta.$$

Iterating (7.4) cancels this factorial:

$$(7.5) \quad I_r(a_r) = \frac{1}{r!} \kappa^r w = (-1)^{r(r-1)/2} \left(\bigwedge^r j_{\mathbb{R}} \otimes \text{id} \right) w.$$

This calculation uses only the staircase equations and its initial coefficient sum. It therefore holds for every collection in the statement.

Set $c = (-1)^{r(r-1)/2} a_r$. Its boundary is a finite sum of differences $(T_{\gamma_i} - 1)b$, so it is closed in $(C_{\bullet, r})_{\Gamma}$. Proposition 6.4 realizes it as a finite rational signed straight tropical cycle Z . The refinement preserves every signed occurrence, all coefficients, and the ordinary marked class. In particular, integration of the constant form gives $I_r(Z) = I_r(c)$, where the integral on Z is summed over one lifted representative of each cell orbit.

We identify this integral with every coordinate of the marked class. Let $\Phi : \mathbb{R}^d \rightarrow V_{\mathbb{R}}$ send the i th standard basis vector to $j(\gamma_i)$, and set $y = \Phi^{-1}x$. The map Φ^{-1} identifies X with $\mathbb{R}^d / \mathbb{Z}^d$. For an increasing r -tuple I , put $dy_I = dy_{i_1} \wedge \cdots \wedge dy_{i_r}$, and choose a rational basis $\{\beta_{\nu}\}$ of B

with dual basis $\{\beta_\nu^*\}$. The positively oriented coordinate r -subtori give the homology basis γ_I , and the forms dy_I give its dual: their integral on the corresponding coordinate subtorus is one and on every other one is zero. This basis also follows from the product cell structure of d circles, whose cellular differentials vanish. Thus, if

$$\text{cl}(Z) = \sum_{I,\nu} c_{I,\nu} \gamma_I \otimes \beta_\nu,$$

then

$$c_{I,\nu} = \int_Z dy_I \otimes \beta_\nu^*.$$

The right-hand sides are exactly the coordinates of $I_r(Z)$ after applying $\wedge^r \Phi^{-1}$ to its vector factor and β_ν^* to its retained coefficient. By (7.5), they are the coordinates of w . Hence $\text{cl}(Z) = w$ in the full marked tensor product, as claimed. $\square$

Theorem 7.2. *For every rank $d \geq 1$ integral tangent lattice N , every full-rank real period lattice $\Gamma \subset V_{\mathbb{R}}$, and every $0 \leq r \leq d$, the complete marked homology classes of finite rational signed straight tropical r -cycles on $X = V_{\mathbb{R}}/\Gamma$ are exactly the rational kernel of the real-target eigenwave. For $1 \leq r \leq d$ this is precisely $\ker W_r$ from Definition 2.5. Every class in this kernel has a representative with lifted vertices in $A = S \otimes V$ for every parameter space S in Set-up 2.1. It has finitely many cell orbits, bounded convex rational-faced cells, rational signed weights, primitive tangent framing, and ordinary balancing, on the original torus and in the exact given rational class. In degree zero the classes are represented by rational weighted points.*

In particular, allowing arbitrary real offsets gives the same class image as requiring vertices in any one of the parameter spaces of Set-up 2.1.

Proof. We compare the two eigenwave targets, prove necessity cell by cell, and apply Proposition 7.1 for sufficiency.

Step 1. We identify the kernel of the parameter-valued eigenwave with the kernel of its real-target counterpart. Suppose first that $1 \leq r \leq d$, and view rational tangent coefficients in their real extensions. Define the real-linear map

$$\mathcal{T}_r : \bigwedge_{\mathbb{R}}^r V_{\mathbb{R}} \otimes_{\mathbb{R}} \bigwedge_{\mathbb{R}}^r V_{\mathbb{R}} \longrightarrow \bigwedge_{\mathbb{R}}^{r-1} V_{\mathbb{R}} \otimes_{\mathbb{R}} \bigwedge_{\mathbb{R}}^{r+1} V_{\mathbb{R}}$$

by

$$\mathcal{T}_r((u_1 \wedge \cdots \wedge u_r) \otimes \beta) = \sum_{a=1}^r (-1)^{a-1} (u_1 \wedge \cdots \widehat{u_a} \cdots \wedge u_r) \otimes (u_a \wedge \beta).$$

The formula is alternating in the u_a , so it defines the displayed map on the exterior power. Under the real period isomorphism $j_{\mathbb{R}}$, this is the real-target eigenwave of Definition 2.5.

The evaluation map on the parameter-valued target is the rational-linear map

$$\begin{aligned} \iota_r : \bigwedge_{\mathbb{R}}^{r-1} G \otimes S \otimes \bigwedge_{\mathbb{R}}^{r+1} V &\longrightarrow \bigwedge_{\mathbb{R}}^{r-1} V_{\mathbb{R}} \otimes_{\mathbb{R}} \bigwedge_{\mathbb{R}}^{r+1} V_{\mathbb{R}}, \\ \zeta \otimes s \otimes \delta &\longmapsto \left(\bigwedge_{\mathbb{R}}^{r-1} j_{\mathbb{R}} \right) (\zeta) \otimes s \delta. \end{aligned}$$

It is injective. To check this, use the basis γ_K of $\bigwedge^{r-1} G$ coming from an integral period basis and the basis v_J of $\bigwedge^{r+1} V$ coming from the fixed tangent basis. Every element of the source is uniquely

$$\sum_{K,J} \gamma_K \otimes s_{K,J} \otimes v_J, \quad s_{K,J} \in S.$$

Its image has real coordinates $s_{K,J}$ in the basis $\{(\bigwedge^{r-1} j_{\mathbb{R}})(\gamma_K) \otimes v_J\}_{K,J}$. They vanish exactly when every $s_{K,J} = 0$ in $S \subset \mathbb{R}$. For $r = d$ both target spaces are zero and the assertion has

the same interpretation. Direct substitution on decomposable tensors gives, for every rational marked class w ,

$$(7.6) \quad \mathcal{T}_r((\bigwedge^r j_{\mathbb{R}} \otimes \text{id})w) = \iota_r(W_r w).$$

Thus the two eigenwaves have the same kernel on the rational marked group. The injection just used is a coordinatewise application of $S \subset \mathbb{R}$ with fixed exterior bases.

Step 2. We show that every rational signed straight cycle, with arbitrary real offsets, has class in the eigenwave kernel. Let Z be a finite rational signed straight tropical r -cycle, with arbitrary real affine offsets. By Proposition 6.4, its primitive-framed chain represents its ordinary marked rational class. Integrate over one bounded lifted representative of each top-cell orbit, retaining its coefficient. Translation leaves these constant-form integrals unchanged. The coordinate-form calculation in the proof of Proposition 7.1 applies to every such closed chain and gives

$$I_r(Z) = (\bigwedge^r j_{\mathbb{R}} \otimes \text{id}) \text{cl}(Z).$$

For one oriented top cell P , choose an oriented basis $u_1, \dots, u_r$ of its real tangent space. Its vector integral is a real multiple of $u_1 \wedge \dots \wedge u_r$. Its coefficient β_P lies in the same tangent determinant line, since it is the weight times the primitive tangent frame. Hence $u_a \wedge \beta_P = 0$ for each a . Every summand in $\mathcal{T}_r(I_r(P \otimes \beta_P))$ is therefore zero. Summing over the finitely many cells gives $\mathcal{T}_r(I_r(Z)) = 0$. Equation (7.6) and injectivity of ι_r now imply $W_r \text{cl}(Z) = 0$. This argument imposes no condition on the vertex positions.

Step 3. We realize every class in the eigenwave kernel over the chosen parameter space and include the degree-zero case. Fix any S in Set-up 2.1. For $w \in \ker W_r$, Proposition 7.1 constructs a finite closed coinvariant chain with vertices in A and exact marked class w . Proposition 6.4 realizes it with the stated bounded rational-faced cells, finitely many cell orbits, rational signed weights, primitive framing, and ordinary balancing. All its identifications use Γ , and the class calculation already gives w itself. This proves sufficiency for every permitted S , including $r = d$.

Finally, for $r = 0$ the coefficient space is $\mathbb{Q}$, the torus is connected, and $H_0(X; \mathbb{Q}) = \mathbb{Q}$. The eigenwave is zero, and a point at the image of $0 \in A$ with any prescribed rational weight represents the corresponding class. This proves the theorem in every degree. $\square$

Corollary 7.3. *For every tropical abelian variety of every dimension and every cycle dimension, each rational tropical Hodge class is the class of a finite rational signed straight tropical cycle.*

The conclusion allows signed rational weights in every cycle dimension.

Proof. By Definition 2.6, a tropical abelian variety has an underlying translation torus as in Set-up 2.1. Its rational tropical Hodge classes are the kernel specified in Definition 2.4. Theorem 7.2 applies to that torus and represents each of those classes. $\square$

8. CONSEQUENCES FOR TROPICAL NONREPRESENTABILITY TESTS

We retain the notation of Set-up 2.1. We explain how Theorem 7.2 excludes a proposed obstruction to the rational Hodge conjecture: a rational tropical Hodge class on a flat torus which has no rational signed balanced representative. We first describe the necessary condition and the flag construction recorded by Zharkov. We then compare the degeneration restrictions of Brosnan and the algebraicity results of Markman with the conclusion of our theorem. The earlier positive tropical results and their historical context are discussed in the introduction.

8.1. The necessary tropical condition. For $0 \leq r \leq d$, write $\mathcal{Z}_r(X; \mathbb{Q})$ for the rational space of finite signed straight tropical cycles of Definition 2.4, and put

$$\mathcal{H}_r(X) = \bigwedge^r G \otimes \bigwedge^r V, \quad \mathcal{K}_r(X) = \ker(\text{real-target eigenwave on } \mathcal{H}_r(X)).$$

In degree zero the eigenvalue is zero. For positive r , Theorem 7.2 identifies $\mathcal{K}_r(X)$ with $\ker W_r$. The class map retains both exterior factors. In Zharkov's notation the period lattice is Γ_1 and the tangent lattice is Γ_2 ; these are our Γ and N , respectively. His tautological framing attaches the integral tangent exterior generator to each oriented cell [Zha20, p. 1]. Proposition 6.4 identifies this rational straight-cycle class with cl .

The strategy is attributed by Zharkov to Kontsevich, with a reference to his January 2012 talk at the conference for Fedor Bogomolov's sixty-fifth birthday [Zha20, pp. 1–2 and bibliography entry 2]. Its force comes from a necessary consequence of algebraicity. To state that consequence precisely, suppose a degeneration of complex abelian d -folds has the identified d -dimensional tropical torus X . Suppose a rational classical Hodge class α of codimension $d - r$ has an identified tropical component $w \in \mathcal{K}_r(X)$. Assume the comparison is defined on the classes under consideration. Assume further that, if α is algebraic in the applicable generic-fiber or spreading setting, a rational algebraic representative spreads after any required finite base change and specializes to a cycle in $\mathcal{Z}_r(X; \mathbb{Q})$. Require this cycle specialization to be rational-linear and compatible with the comparison. More explicitly, if σ denotes the chosen map on classes and trop denotes this cycle specialization, the required compatibility is

$$\sigma(\text{cl}_{\text{alg}}(Z_{\text{alg}})) = \text{cl}(\text{trop}(Z_{\text{alg}})), \quad \sigma(\alpha) = w.$$

Here cl_{alg} is the classical cycle-class map in degree $2(d - r)$, with the homology–cohomology identification specified by the chosen comparison. These assumptions include any base change and spreading needed to define the specialization and its marking. They are assumptions on the particular degeneration, rather than a comparison theorem for every abstract flat torus.

Under these assumptions, an algebraic representative of α specializes to a tropical representative of w . Thus the proposed obstruction requires

$$(8.1) \quad w \in \mathcal{K}_r(X), \quad w \notin \text{cl}(\mathcal{Z}_r(X; \mathbb{Q})).$$

Such a w , together with the comparison, would obstruct algebraicity of α . The argument neither assigns a classical Hodge class to every abstract tropical class nor constructs an algebraic lift of a tropical representative. It makes explicit the compatibility needed to use the degeneration argument described in [Zha20, p. 1].

Rational coefficients and signed weights are essential to this test. A rational algebraic cycle has the form

$$Z_{\text{alg}} = \sum_{i=1}^m a_i [Y_i], \quad a_i \in \mathbb{Q},$$

where the Y_i are subvarieties of the required dimension. A linear specialization carries these coefficients, including their signs, to the tropical cycle. Failure to find an effective representative does not exclude such a sum. Nor does failure to find an integral representative of one normalization exclude a rational representative. By Proposition 6.4, a common positive denominator turns any cycle in $\mathcal{Z}_r(X; \mathbb{Q})$ into an integer straight cycle, and division by that denominator recovers its exact rational class. Balancing applies to the total signed cycle; its positive and negative parts need not separately balance.

8.2. Zharkov's flags and the proposed separating functional. For this description take $d = 4$ and $r = 2$. Zharkov gives an explicit four-parameter family of tropical fourfolds with imaginary quadratic symmetry and two additional rational Hodge classes w_1, w_2 [Zha20, p. 2]. Let ϑ denote the polarization-square class in that description. These three classes provide distinguished directions on which to test an obstruction. The proposed certificate is a linear identity for all balanced cycles, rather than a search through a finite list of cycle supports.

We describe the two maps in the proposal, following [Zha20, pp. 3–4]. Let Π be its rank-four formal parameter lattice and put $\Pi_{\mathbb{Q}} = \Pi \otimes_{\mathbb{Z}} \mathbb{Q}$. The formal period marking places period vectors in $\Pi_{\mathbb{Q}} \otimes V$. Prospective polygonal chains have vertices in this parameter space modulo the marked periods, and rational edge directions. Zharkov organizes the construction using triangles

and parallelograms, allowing finite denominators in the vertex data when subdividing. This is the rational parameter framework of his proposal; it is not a restriction on the periods in Theorem 7.2.

An affine flag records a vertex x , a rational edge line ℓ , and the tangent bivector β of an incident two-dimensional face. At fixed (x, ℓ) the flag space is linear in β . Denote this space by $\mathcal{F}$, the polygonal chain space by $\mathcal{P}$, and the map taking a polygon to its signed incidence flags by

$$\mathbf{b} : \mathcal{P} \longrightarrow \mathcal{F}.$$

This is the map denoted α by Zharkov. Its relevance is the identity $\mathbf{b}(Z) = 0$ for a balanced cycle: the retained bivector boundary contributions cancel at each edge, and hence their endpoint flags cancel. The linearity in the face coefficient is what allows distinct incident planes to enter the same balancing equation.

The other map records symbolic area with its tangent framing. Its target is

$$\mathcal{V} = \text{Sym}^2(\Pi_{\mathbb{Q}}) \otimes \text{Sym}^2\left(\bigwedge^2 V\right).$$

Write $\text{vol}_{\text{sym}} : \mathcal{P} \rightarrow \mathcal{V}$ for this map. If $\mathcal{R} \subset \mathcal{V}$ is a rational subspace and $q_{\mathcal{R}} : \mathcal{V} \rightarrow \mathcal{V}/\mathcal{R}$ is the quotient, the sought map $\Phi : \mathcal{F} \rightarrow \mathcal{V}/\mathcal{R}$ would satisfy

$$\Phi \circ \mathbf{b} = q_{\mathcal{R}} \circ \text{vol}_{\text{sym}}.$$

It would follow that $q_{\mathcal{R}}(\text{vol}_{\text{sym}}(Z)) = 0$ for every balanced cycle in this presentation. To obtain a nonrepresentability certificate, the construction must cover all cycles relevant to the chosen class, the resulting invariant must depend only on that class, and a proposed Hodge class must have nonzero image in the quotient.

The elementary identities show what this demand entails. Write $\Phi_{x,u}$ for the linear map on $\bigwedge^2 V$ at the vertex x and rational line spanned by u . For independent integral tangent vectors u, v and parameter values $s, t \in \Pi_{\mathbb{Q}}$, the triangle and parallelogram equations of [Zha20, p. 4, equations (1)–(2)] take the following form after passing to the proposed quotient:

$$\begin{aligned} & (\Phi_{x,u} - \Phi_{x,v} + \Phi_{x+su,u-v} - \Phi_{x+su,u} + \Phi_{x+sv,v} - \Phi_{x+sv,u-v})(u \wedge v) \\ & \quad = q_{\mathcal{R}}(s^2 \otimes (u \wedge v)^2), \\ & (\Phi_{x,u} - \Phi_{x,v} + \Phi_{x+su,v} - \Phi_{x+su,u} - \Phi_{x+tv,u} + \Phi_{x+tv,v} \\ & \quad + \Phi_{x+su+tv,u} - \Phi_{x+su+tv,v})(u \wedge v) \\ & \quad = q_{\mathcal{R}}(2st \otimes (u \wedge v)^2). \end{aligned}$$

Here s^2, st are symmetric products in the formal parameter space, and $(u \wedge v)^2$ is a symmetric square. The maps must also be periodic in x for the marked period lattice. The cell equations and this periodicity must hold together for a functional on the torus.

The symbolic target has a separate normalization. After evaluating s, t as real lengths, the triangle with edge vectors su, sv has ordinary oriented vector area $s^2(u \wedge v)/2$, and the parallelogram with edge vectors su, tv has area $st(u \wedge v)$. With exterior coefficient $u \wedge v$, the two right-hand sides above use twice these areas and then the symmetric product of the two bivectors. Thus this symbolic volume is not the full marked tensor in $\bigwedge^2 G \otimes \bigwedge^2 V$. A class-level detector constructed from it must include the period expansion, symmetrization, and this normalization. Equality of symbolic images alone is not equality of complete marked classes.

Zharkov's final ansatz imposes linearity in s and u on the differences

$$\Delta_{x,s,u} = \Phi_{x+su,u} - \Phi_{x,u}.$$

Together with the cocycle condition, the ansatz imposes $\Delta_{x+tu,s,u} = \Delta_{x,s,u}$. Periodicity then leads to a finite linear system. Zharkov reports that the ansatz solves the equations modulo the entire lattice generated by ϑ, w_1, w_2 , but does not give the desired solution modulo a proper sublattice [Zha20, p. 4, final paragraph]. It therefore supplies no separated Hodge class and

no counterexample. For a rational obstruction, even a proper sublattice would have to remain proper after taking rational spans: a finite-index discrepancy disappears over $\mathbb{Q}$. Quotienting out the full rational span of the three candidate directions leaves none of them available for detection.

8.3. Weil type, maximal degeneration, and the classical exclusions. The source of the candidate classes persists in higher dimensions. Let $n \geq 1$ and let Y be a complex abelian variety of dimension $2n$ with an action of an imaginary quadratic field K_0 . It is of Weil type when the two embeddings of K_0 occur with multiplicities (n, n) on $H^{1,0}(Y)$. After extending scalars to K_0 , write U_+, U_- for the two eigenspaces in $H^1(Y; \mathbb{Q}) \otimes K_0$. Each has dimension $2n$ over K_0 , and

$$\bigwedge_{K_0}^{2n} U_+ \oplus \bigwedge_{K_0}^{2n} U_- \subset H^{2n}(Y; K_0)$$

descends to the two-dimensional rational space of Weil classes. Each determinant line has Hodge type (n, n) , because its eigenspace has n directions of each Hodge type. This description and the polarized Weil-type conventions are recalled in [Mar25, §1.1, p. 3]. It explains why these families supply Hodge classes in addition to powers of a polarization.

The possible maximal degenerations impose an arithmetic restriction. A Weil-type polarization determines a nondegenerate K_0 -Hermitian form $\mathfrak{h}$ on $H_1(Y; \mathbb{Q}) \cong K_0^{2n}$, of signature (n, n) . If $B_{\mathfrak{h}}$ is its matrix in a K_0 -basis, we use the determinant convention

$$\text{disc}(\mathfrak{h}) = [\det B_{\mathfrak{h}}] \in \mathbb{Q}^{\times} / \text{Nm}_{K_0/\mathbb{Q}}(K_0^{\times}).$$

A basis change multiplies the determinant by a norm. The orthogonal sum of n Hermitian planes with matrix $\text{diag}(1, -1)$ has discriminant $[(-1)^n]$ in this convention. An additional sign normalization can call this the trivial discriminant; it does not make the determinant class equal to $[1]$ when n is odd. These are the conventions of [Bro26, §4, p. 5] and [Mar25, §1.1].

We specify the maximality condition used by Brosnan. Consider a smooth projective family $\mathcal{Y} \rightarrow B$ of polarized Weil-type abelian $2n$ -folds, with its quadratic-field action and Weil-type polarization. Here B is a smooth complex curve contained in a smooth connected curve $\overline{B}$ with $\overline{B} \setminus B = \{b_0\}$, and the associated Hermitian form on a fiber is isometric to $\mathfrak{h}$. Fix $b \in B$ and let $\mathcal{N}$ be the logarithm of the unipotent part of the monodromy about b_0 on $H_1(\mathcal{Y}_b; \mathbb{Q})$. Maximality means

$$\mathcal{N}^2 = 0, \quad \text{rank}_{\mathbb{Q}} \mathcal{N} = 2n.$$

Thus, on a rational space of dimension $4n$, there are $2n$ Jordan blocks of size two. The square-zero assertion holds for this weight-one degeneration; the rank condition says that half the one-cycles vanish. It is the condition used to obtain a tropical torus of real dimension $2n$ from abelian varieties of complex dimension $2n$ [Bro26, §4, p. 5].

Brosnan's Proposition 4 states that such a maximal degeneration implies the existence of an n -dimensional totally isotropic K_0 -subspace. Existence of that subspace is equivalent to signature (n, n) and determinant discriminant $[(-1)^n]$ [Bro26, Proposition 4, pp. 4–5]. The key calculation in his proof explains the restriction. The logarithm $\mathcal{N}$ is K_0 -linear and skew-adjoint for $\mathfrak{h}$. Consequently $W = \text{im } \mathcal{N}$ has dimension n over K_0 and

$$\mathfrak{h}(\mathcal{N}x, \mathcal{N}y) = -\mathfrak{h}(x, \mathcal{N}^2y) = 0.$$

It is a maximal totally isotropic subspace [Bro26, §6, pp. 7–8]. We use the implication from maximal degeneration to this subspace and the stated Hermitian equivalence. No converse existence theorem for degenerations is needed here.

In dimensions four and six this necessary discriminant is precisely the one occurring in the following algebraicity results. Markman's published Theorem 1.5, identified there with Theorem 13.4, proves that the three-dimensional space generated by the polarization square and the Weil classes of a polarized Weil-type abelian fourfold of discriminant $[1]$ consists of algebraic classes [Mar23, Theorem 1.5 (= Theorem 13.4), p. 236]. This applies to those distinguished

classes on every such fourfold; the full Hodge-conjecture consequence stated in that theorem is for the generic such fourfold.

The later Theorem 1.5.1 of Markman proves algebraicity of the Hodge–Weil classes on polarized Weil-type sixfolds of discriminant $[-1]$, for every imaginary quadratic field [Mar25, Theorem 1.5.1, p. 9]. This is a statement about the Weil classes in that family. His Corollary 1.6.1 proves the rational Hodge conjecture for all abelian fourfolds [Mar25, Corollary 1.6.1, p. 9]; its proof combines Schoen’s degeneration argument with the reductions recalled there. The sixfold theorem is not a statement about all Hodge classes on all abelian sixfolds.

Combining the stated results gives the following comparison. Brackets in the middle column denote determinant classes modulo norms.

Complex mension	di- nant	Required discrimi- nant	Role of the quoted algebraicity results
4		$[1]$	The relevant Weil classes are algebraic by the published fourfold theorem; the later corollary covers all fourfolds.
6		$[-1]$	The sixfold theorem makes the targeted Weil classes algebraic in the component allowed by maximality.
8		$[1]$	The quoted fourfold and sixfold statements do not supply an algebraicity conclusion in this dimension.
$2n \geq 10$		$[(-1)^n]$	The same degeneration restriction persists, without an all-dimensional algebraicity conclusion from those statements.

For the fourfold and sixfold Weil classes in question, changing the discriminant therefore does not restore the proposed maximal degeneration test: the permitted discriminant is covered by the algebraicity result, and the other discriminants fail the necessary degeneration condition. This is the comparison discussed by Brosnan [Bro26, §1 and §7]. We use Proposition 4 for its precise direction and discriminant condition. His degeneration restriction and the tropical realization theorem address different requirements of the strategy.

Brosnan records that the first version of his note was written in April 2023, when the published fourfold result supplied the relevant algebraicity input. The 2026 preprint also incorporates the comparison with Markman’s later sixfold result [Bro26, §1 and §1.1].

The low-dimensional comparison by itself does not give a uniform answer as n increases. Markman explicitly notes, immediately after Theorem 1.5.1, that the semiregularity proof used there is specific to genus three, although the construction of the objects extends further [Mar25, p. 9]. Thus those particular algebraicity theorems alone leave a higher-dimensional part of the proposed test unsettled. This observation concerns the scope of the quoted results; it is not a classification of subsequent classical Hodge-theoretic results.

8.4. Exclusion in every dimension and every marked class. Theorem 7.2 settles the tropical nonrepresentability condition itself. In the notation above, it gives

$$(8.2) \quad \text{cl}(\mathcal{Z}_r(X; \mathbb{Q})) = \mathcal{K}_r(X) \quad (0 \leq r \leq d)$$

for every rank $d \geq 1$ tangent lattice and every full-rank real period lattice. Equation (8.2) contradicts the second requirement in (8.1) whenever the first holds. This conclusion does not require a classical algebraicity theorem in each dimension.

The order of the quantifiers matters. One may first fix the dimension, the integral tangent lattice, the actual period lattice, the cycle dimension, and any rational class in the real-target eigenwave kernel. The theorem then supplies a representative of that class on that torus. In the Weil-type setting this applies with $d = 2n$ and $r = n$ to every identified tropical Weil direction which satisfies the eigenwave condition, and to every rational combination of those directions. It

also applies to every other rational tropical Hodge class. Increasing n , changing the imaginary quadratic field, or selecting a different Hodge direction cannot produce a witness of (8.1).

The period quantifier includes arbitrary real period coordinates. There is no rational-period, genericity, or rational-triangulability assumption. A polarization, when present, adds no restriction needed by the proof; changing it or changing an allowable Weil discriminant does not alter the implication. No condition excluding rational subtori occurs. Finally, for every parameter space S allowed in Set-up 2.1, the representative can be chosen with vertices in A . The rational coefficients do not require the real affine offsets to be rational numbers.

The geometric and marked conclusions are also necessary for this application. The representative has finitely many cell orbits, bounded convex rational-faced cells, ordinary primitive-normal balancing, and the primitive tangent framing of the class map. It therefore belongs to the cycle category whose nonrepresentability was to be tested. Proposition 6.4 preserves the actual occurrence multiplicities and the original period identifications. In particular the conclusion is on $V_{\mathbb{R}}/\Gamma$ with the given marking.

The complete class computation supplies the required equality before any symbolic symmetrization. For $1 \leq r \leq d$, the staircase $a_0, \dots, a_r$ of Proposition 7.1 satisfies

$$I_r(a_r) = (-1)^{r(r-1)/2} \left(\bigwedge^r j_{\mathbb{R}} \otimes \text{id} \right) w.$$

The refined periodization of $(-1)^{r(r-1)/2} a_r$ consequently has marked class exactly w . There is no residual factorial or lattice index, and no replacement of the deck lattice by a finite cover. This full equality addresses the chosen witness even when a symmetrized volume would discard some of its coordinates. The construction is finite, so it also gives an actual cycle rather than an approximation or a limit of cycle classes.

The same conclusion rules out every linear class detector of the kind sought in the flag proposal.

Corollary 8.1. *For every torus and every cycle dimension in Theorem 7.2, let $H = \bigwedge^r G \otimes \bigwedge^r V$, and let $K \subset H$ be its rational real-target eigenwave kernel. If D is a rational vector space and $L : H \rightarrow D$ is a rational linear map which vanishes on the classes of all finite rational signed straight tropical r -cycles, then $L|_K = 0$.*

Proof. Let $w \in K$. By Theorem 7.2 there is a finite rational signed straight tropical r -cycle Z on the given torus with $\text{cl}(Z) = w$. The hypothesis on L gives $L(w) = L(\text{cl}(Z)) = 0$. This holds for every $w \in K$. $\square$

The target D may itself be a quotient space. Thus the corollary covers a flag functional whenever its output descends to a rational linear invariant of the complete marked class and vanishes on every cycle. If such a class invariant takes a nonzero value on some $w \in K$, the theorem supplies a genuine cycle with that same value, so the invariant cannot vanish on all cycles. Restricting the polygonal ansatz cannot turn this into a universal nonrepresentability certificate. More generally, even without linearity, a representative-independent test cannot correctly certify the absence of ordinary representatives for a class for which the theorem constructs one. The corollary gives the precise linear formulation; the absence of the witness follows directly from (8.2).

8.5. Representability and algebraic liftability. For a classical class α and tropical component w in the comparison considered above, there are three distinct requirements:

- (i) w belongs to the rational real-target eigenwave kernel.
- (ii) w is represented by an ordinary finite rational signed balanced straight tropical cycle on the given torus.
- (iii) A representative of w is the compatible specialization of a rational algebraic cycle representing α in the prescribed degeneration or spreading setting.

The substantive sufficiency proved here is (i) $\Rightarrow$ (ii). The necessary eigenwave condition gives (ii) $\Rightarrow$ (i), and the assumed compatibility of specialization gives (iii) $\Rightarrow$ (ii). No implication (ii) $\Rightarrow$ (iii)

is proved. In particular, the existence of the tropical representative does not establish algebraicity of α .

Effectivity and integrality impose different requirements from (ii). The theorem does not prescribe nonnegative weights or an integral representative of the exact integral class. A finer obstruction might also retain a phase, logarithmic structure, deformation compatibility, or information about the full limiting class. Such data are not part of the ordinary marked tropical class considered here. To use them against algebraicity would require a necessary condition for specializations of algebraic cycles and a violation by the same chosen rational classical class. No such violation follows from this paper.

A partial degeneration can retain an abelian part as well as a tropical torus. Brosnan mentions less-than-maximal degenerations as a possible different setting [Bro26, §7]. Our theorem controls the ordinary cycle-class image on any flat translation torus that occurs; it does not identify the specialization image including the remaining abelian part or mixed data. Likewise, general tropical spaces with singularities, or integral affine spaces with nontrivial linear holonomy, are outside the stated hypotheses. These distinctions mark the scope of the result. The excluded obstruction is precisely failure of rational signed tropical representability in the flat-torus eigenwave kernel.

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. We record the configuration and chronology in Section A.1.

A.1. How the system was run. The research execution records identify the base model as `gpt-6-astra`. The main agent used reasoning effort `ultra`. There were seven evidentiary workers: four at `max` and three at `xhigh`. The fact verifier used `max`. Three exploratory helper homes appear in the records before acceptance of the signed-realization theorem; these helpers are separate from the seven-worker roster. The execution records establish these model and effort settings without asserting that all actors ran continuously or simultaneously.

The project was created on 17 September 2026 at 22:04:22 UTC. The retained submission receipt records acceptance of the positive signed-realization theorem, source identifier `cdddcc589e81efad`, on 18 September 2026 at 02:51:06.188 UTC. The elapsed wall-clock interval from project creation to that event was 4 hours, 46 minutes, and 44.188 seconds, or about 4 hours and 47 minutes. Subsequent source-category checks and independent audits took place later. An earlier sign-error cascade was revoked; its earlier claims do not supply the acceptance time reported here. The current source agrees byte for byte with the frozen source selected for this manuscript.

The original experiment sought a counterexample to the rational classical Hodge conjecture through tropical nonrepresentability. It instead produced the positive signed-realization theorem used in this paper. The original counterexample objective was not achieved. Later research formulations asked about algebraic realizability and then a CM-point deformation and algebraization obstruction. Those later formulations remained unfinished and are not results of this manuscript.

No controlled comparison run is established by the inspected records. An instruction to create a separate project does not provide a same-task benchmark, its outcome, or its time to completion. The retained aggregate usage report also includes later research and paper activity, so it does not establish a theorem-specific token or monetary cost.

The paper-only preparation began on 21 September 2026. Six native section-writing jobs produced the proof sections and the detailed consequences section. Each section passed its independent native review, and the assembled mathematical body passed a separate whole-paper review before polish. The controller wrote the outline, introduction, front matter, and disclosure, and integrated the sections under the native writing workflow. Sources and review and build records are retained privately. None of the proofs in the mathematical body depends on a computational experiment requiring a separate published computation appendix.

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